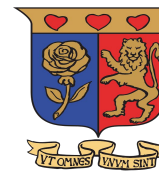


Strathmore
UNIVERSITY

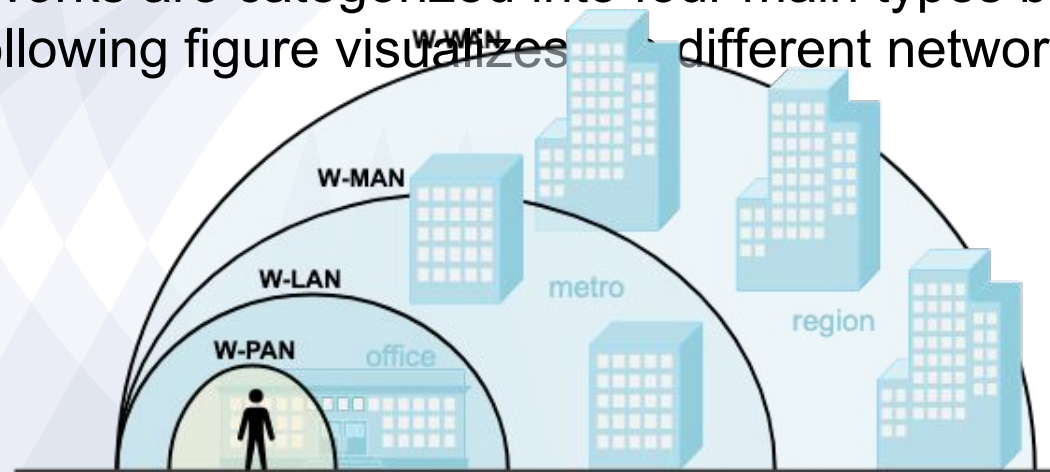
Wireless Networks and WLANs

WLAN Infrastructure



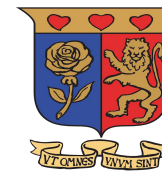
Strathmore
UNIVERSITY

- There is a common misconception among some people that all wireless networks are Wi-Fi. In reality, Wi-Fi is just one type of wireless network. Technologies like Bluetooth, ZigBee, WiMAX, LTE, and 5G are other types, each using RF signals but differing in their specifications and uses. Wi-Fi, specifically, refers to the wireless protocols defined by the IEEE in the 802.11 specification.
- Wireless networks are categorized into four main types based on the geographic area they cover. The following figure visualizes different network types and their coverage areas:



- We use the term "wireless," to refer to Wireless Local Area Networks (WLANs) based on the IEEE 802.11 standard. These are the Wi-Fi networks we encounter daily in homes, offices, cafes, and public spaces. While other technologies exist, we will focus on WLANs, their operation, components, and best practices for designing and troubleshooting them.

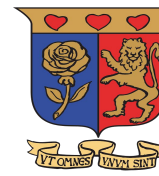
Wireless Network Types



Strathmore
UNIVERSITY

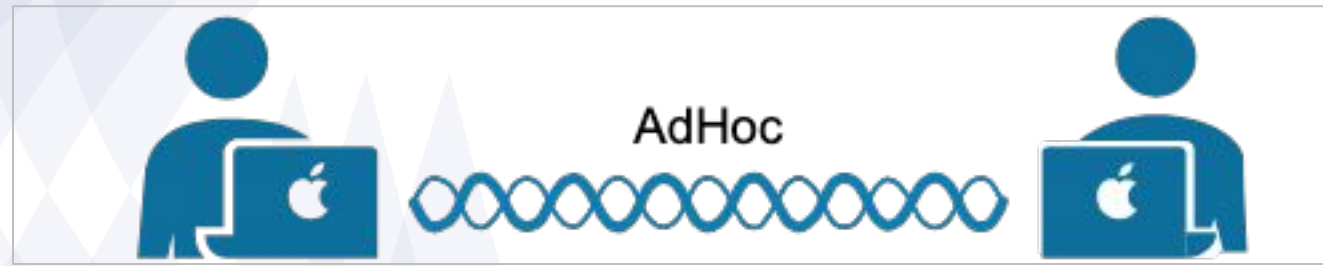
Name	IEEE Standard	Description
WPAN (Wireless Personal-Area Network)	802.15.1	These are personal wireless networks that use very low-power transmitters for short-range connections. Common example is Bluetooth that people use to connect their earbuds, keyboards and mice. Typically, within the range of a few meters.
WLAN (Wireless Local-Area Network)	802.11	The typical wireless network everybody knows - at home, in the office and at the local cafe. It is referred to as Wi-Fi . This is the type of wireless network that we are going to discuss throughout this course. Wi-Fi networks connect multiple devices over a medium range, usually up to 100 meters. They use the <i>IEEE 802.11</i> standard and operate on unlicensed frequencies 2.4Ghz, 5Ghz and 6GHz.
WMAN (Wireless Metropolitan-Area Network)	802.16	These networks provide wireless service over a large area, such as part of a city. WiMAX is a common example, based on the IEEE 802.16 standard. They become less common these days because of the rapid adoption of LTE and 5G.
WWAN (Wireless Wide-Area Network)	LTE/5G	These networks offer wireless data services over very large areas, including regional, national, or even global coverage. They are provided by telecommunications carriers and mobile phone operators.

The Wireless LAN: Ad-Hoc(Peer-to-Peer)



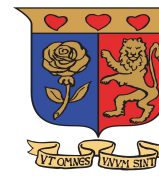
Strathmore
UNIVERSITY

- A wireless local area network (WLAN) can consist of only one device or connect thousands of devices. The most basic form of WLAN is an ad hoc network.
- Ad-hoc is a type of wireless LAN where devices communicate directly with each other without relying on a central access point (AP) or a wireless router. These networks are typically used for temporary connections and are created on the fly, often called "peer-to-peer" networks, as shown in the diagram below.

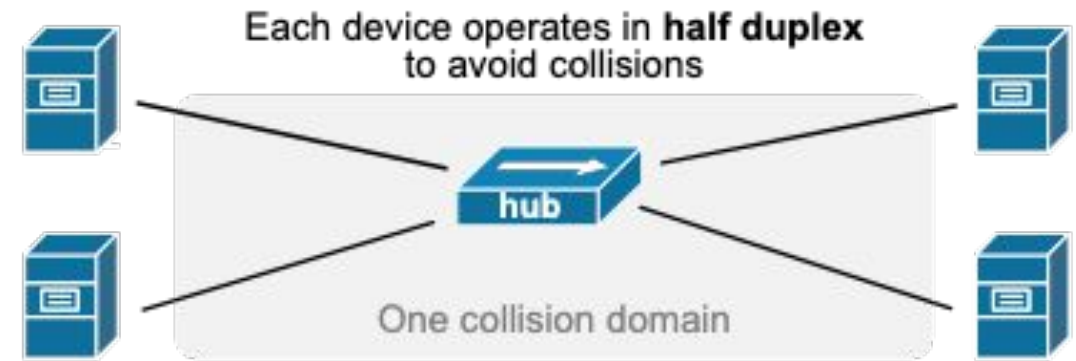
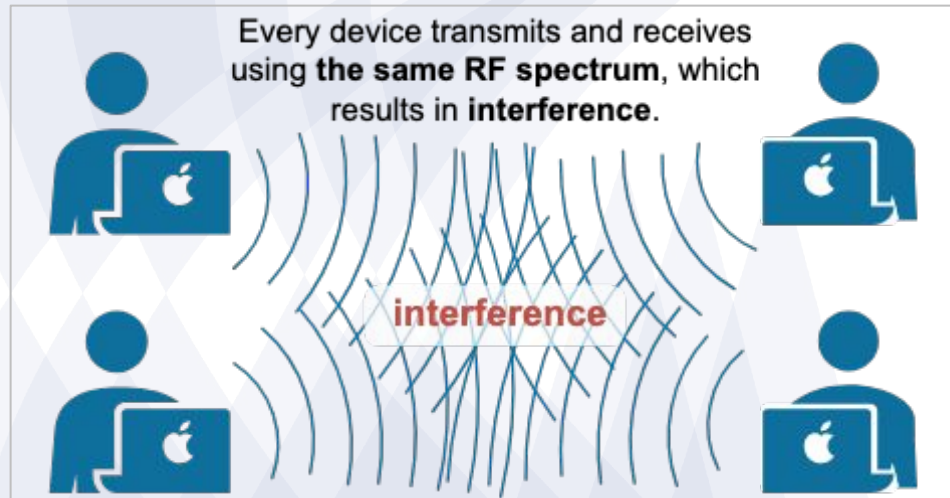


- In an ad hoc network, each node functions as both a client and a router. This setup is useful when wireless infrastructure is unavailable or impractical, such as in spontaneous gatherings outside.
- Ad-hoc networks offer flexibility and ease of deployment, but they may face challenges such as limited range, lower data rates, and potential security vulnerabilities compared to traditional infrastructure-based WLANs.

The Wireless LAN: Ad-Hoc(Peer-to-Peer)

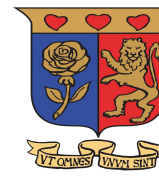


- Additionally, ad-hoc networks don't scale to connecting multiple devices. The reason is that if multiple signals are received at the same time, they interfere with each other, and this interference becomes more likely as more wireless devices are added. For example, the following diagram shows four devices on the same channel and what might happen if some or all of them transmit simultaneously.



- If you think about it, this is the same effect that happens in an Ethernet LAN that is a single collision domain. Recall that all devices connected to a hub are in the same collision domain. To use the media effectively, all hosts must operate in half-duplex mode, which means they can't send and receive data at the same time.

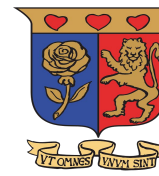
The Wireless LAN: Ad-Hoc(Peer-to-Peer)



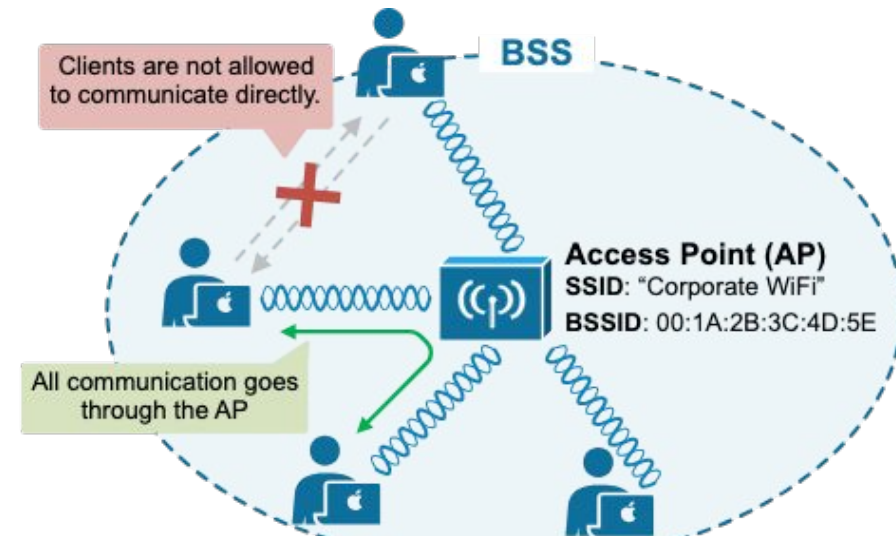
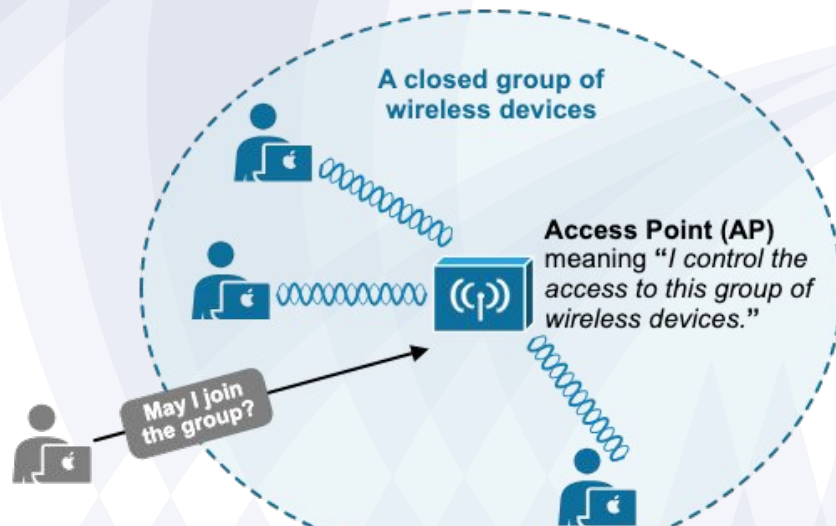
Strathmore
UNIVERSITY

- In a Wi-Fi network, when multiple devices transmit at the same time on the same frequency, their signals collide, making communication impossible. This now introduces the working of a half-duplex communication in wireless networks. A Wi-Fi device can either send or receive data at any given moment, but not at the same time. It must wait for its turn, just like in a conversation.
- In Wi-Fi, this is managed by collision avoidance mechanisms, which help devices sense if the channel is clear before transmitting. So, just like in a conversation where people take turns speaking to avoid confusion, wireless devices must take turns transmitting data to avoid interference and ensure clear communication..
- **Note:** In a wireless LAN, all devices share the same wireless medium—free space—much like people competing for airtime in a conversation. To prevent collisions, only one device can transmit data at a time. Hence, wi-fi communication is half-duplex.
- Now, knowing that all Wi-Fi clients share the same medium (free space), consider the security and privacy aspects of the communication. With peer-to-peer connections, no one can stop a rogue device from transmitting wherever it wants, causing collisions.
- However, at the very least, a wireless network must ensure that all devices on a channel follow the same basic settings, such as data rates, modulation types, and channel width. Additionally, measures should be taken to control which devices and users can access the network and how wireless transmissions are secured.

The Wireless LAN: Basic Service Set(BSS)

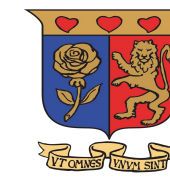


- The solution was to create closed, controlled groups of wireless devices. A central device called an **access point (AP)** manages access to the group and the settings of all devices(graphic 1).

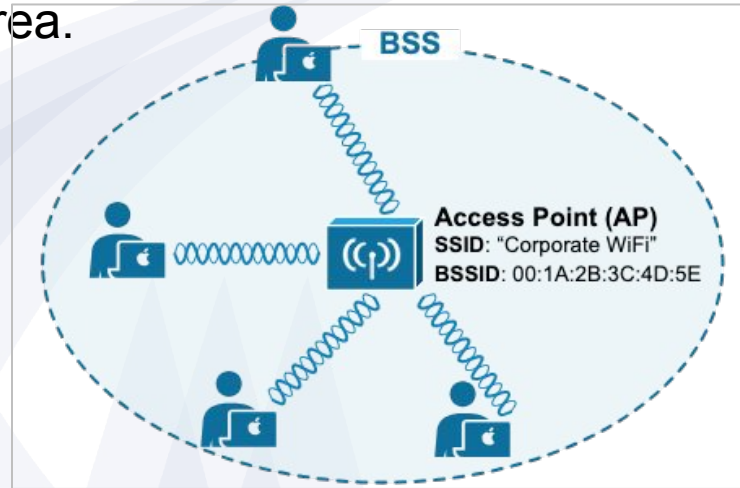


- Before joining the group, a device must announce its capabilities and receive permission from the centralized access point (AP). In the 802.11 standard, this type of network is called a Basic Service Set (BSS). A wireless access point (AP) is at the core of every BSS (graphic 1).
- BSS (Basic Service Set) is the fundamental building block of a Wi-Fi network. It refers to a group of wireless devices that communicate with each other through a single Access Point (AP) (graphic 2).
- Inside a BSS, the AP, which ensures data is transmitted properly using the same frequency spectrum, modulation, channel, and so on. Each BSS is identified by a unique BSSID (Basic Service Set Identifier), which is typically the MAC address of the AP's radio interface.

The Wireless LAN: Basic Service Area (BSA)

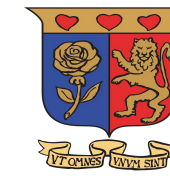


- Since the AP is the heart of a BSS, its network coverage is limited to the area where the AP's signal is available; called a **cell** or the **Basic Service Area (BSA)**. The graphic represents the cell as a circular area where the coverage shape depends on the AP's antenna type. The cell area can be circular if the AP uses an omnidirectional antenna. However, there are APs with different antenna types, which change the shape of the coverage area.

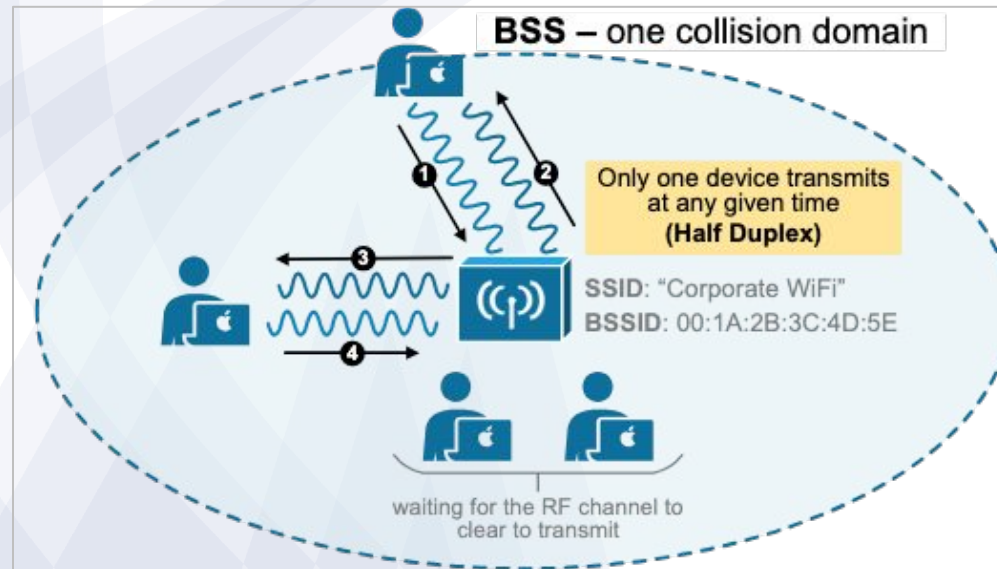


- The AP is the primary contact point for all devices trying to join the network. It broadcasts the presence of the BSS so that devices can discover it and attempt to connect. The AP identifies itself using a Basic Service Set Identifier (BSSID) based on its radio MAC address. The AP also advertises a Service Set Identifier (SSID), a human-readable name for the wireless network.
- Think of the BSSID as a unique machine-readable identifier for the AP, while the SSID is a user-friendly name that helps people recognize the network.

The Wireless LAN: Basic Service Area (BSA)

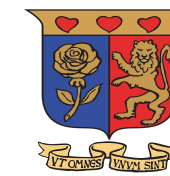


- **Device Association:** Before joining the BSS, a device must request permission from the AP, a process called association. Once accepted, the device becomes a client, also known as an 802.11 station (STA). After associating, most of the client's communication must go through the AP. Data frames are sent or received using the BSSID as the source or destination address.

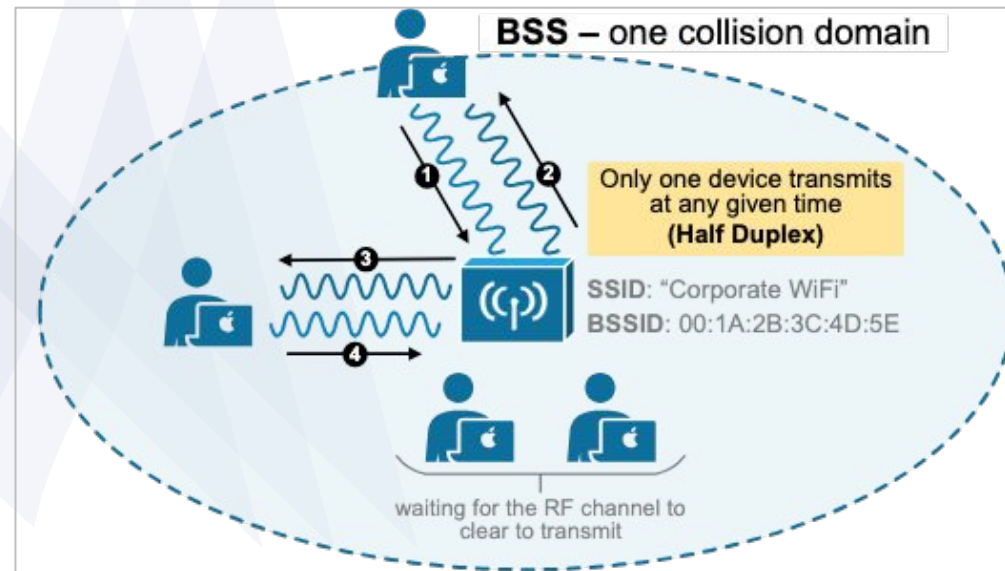


- All communication between devices inside a Basic Service Set (BSS) must go through the Access Point (AP). The AP acts as the central hub that controls and manages all traffic. When a device wants to send data to another device in the same BSS, it first sends the data to the AP, as shown in the diagram below. The AP then forwards the data to the receiving device.

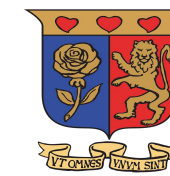
The Wireless LAN: Basic Service Area (BSA)



- You might wonder why two clients in the same BSS cannot just communicate directly. The reason is that routing all traffic through the AP helps maintain order and stability within the network. If clients were to bypass the AP, managing the BSS would become impossible.
- Even though data is sent through the AP, any device in the same area tuned to the same channel can still receive the transmissions. Wireless frames are broadcasted openly, so anyone within range can read their contents if they are not encrypted. The only thing identifying the intended sender or receiver is the BSSID inside the frame.



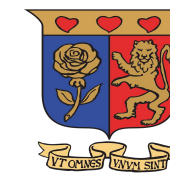
WLAN Infrastructure: Autonomous AP Architecture



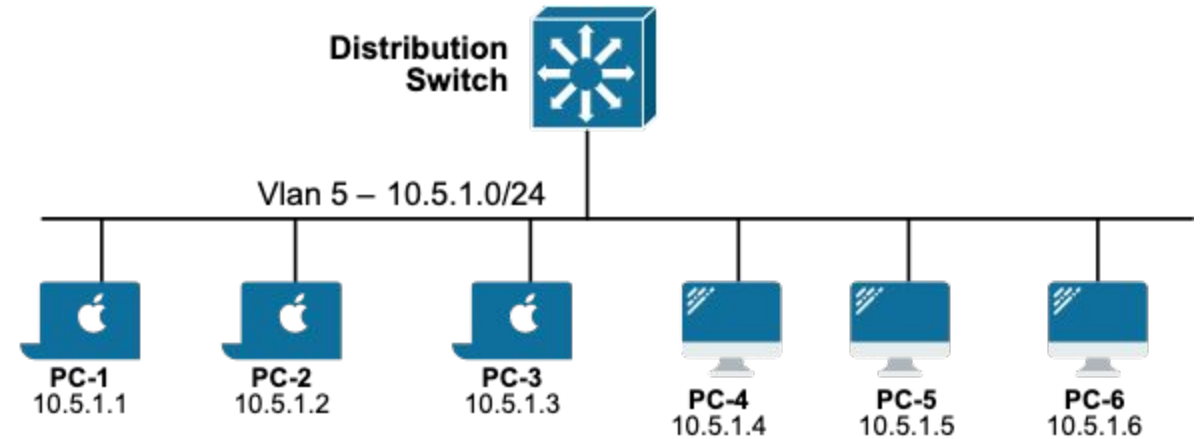
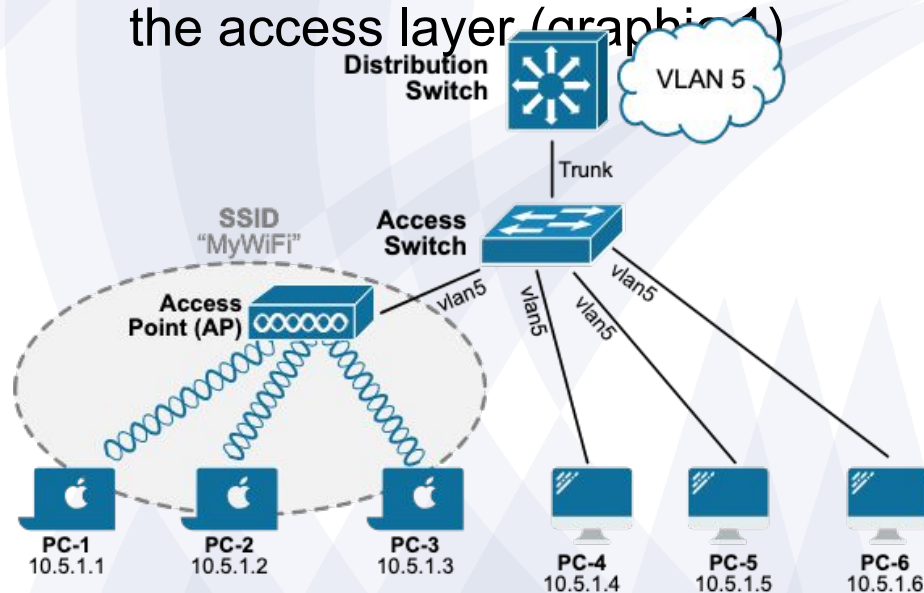
Strathmore
UNIVERSITY

- Based on the size of the enterprise infrastructure and the operational mode of the access points (APs), the wireless network can be designed using three different architecture models:
 - Autonomous AP Architecture
 - Lightweight AP Architecture
 - Cloud-based Architecture
- Each architecture has its advantages and is chosen based on factors like network size, management needs, and budget.
- The access point (AP) allows wireless clients to connect to a wired network using radio signals, just as if they were connected via cables to the switched network. In other words, the AP provides and controls mobile clients' access to the wireless network (hence the name "access point").
- The wireless network that AP creates is called a Basic Service Set (BSS). It consists of an access point (AP) and the wireless devices connected to it, as shown in the diagram below.
- Each BSS operates within a specific coverage area (the dashed line) and is identified by a unique BSSID, usually the AP's MAC address. In a typical corporate network, multiple BSSs can be linked together to form a larger wireless network. This is called ***an Extended Service Set (ESS)***. It consists of multiple BSSs connected through a common wired network. Each BSS has its own access point (AP), but all APs share ***the same Service Set Identifier (SSID)***, allowing seamless roaming for wireless devices.

WLAN Infrastructure: Autonomous AP Architecture

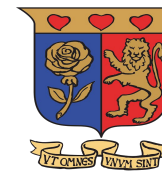


- An Autonomous Access Point (AP) works as a standalone device that provides a fully functional wireless network (BSS). It basically extends the switched network via a wireless signal to mobile devices. It maps a wireless network (SSID) to a wired network (VLAN) at the access layer (graphic 1)

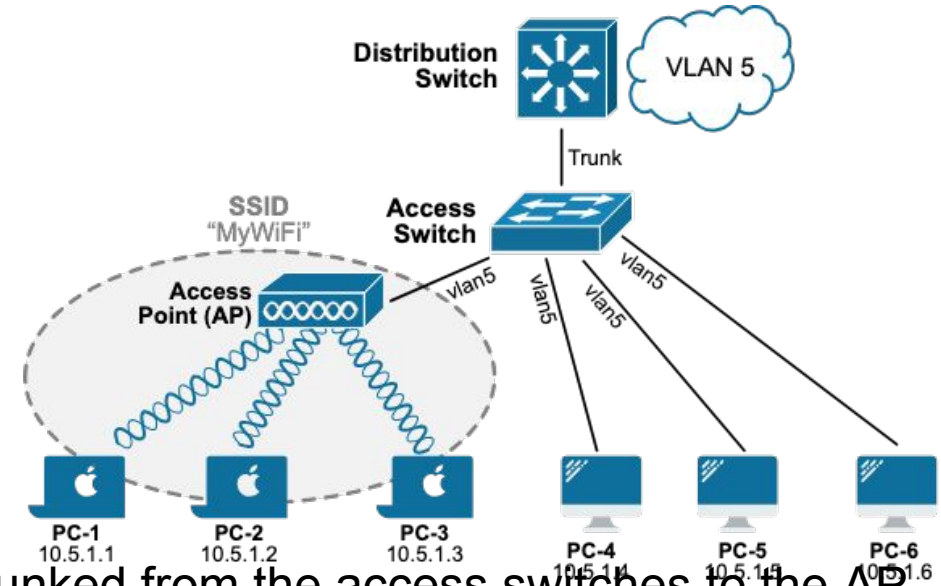
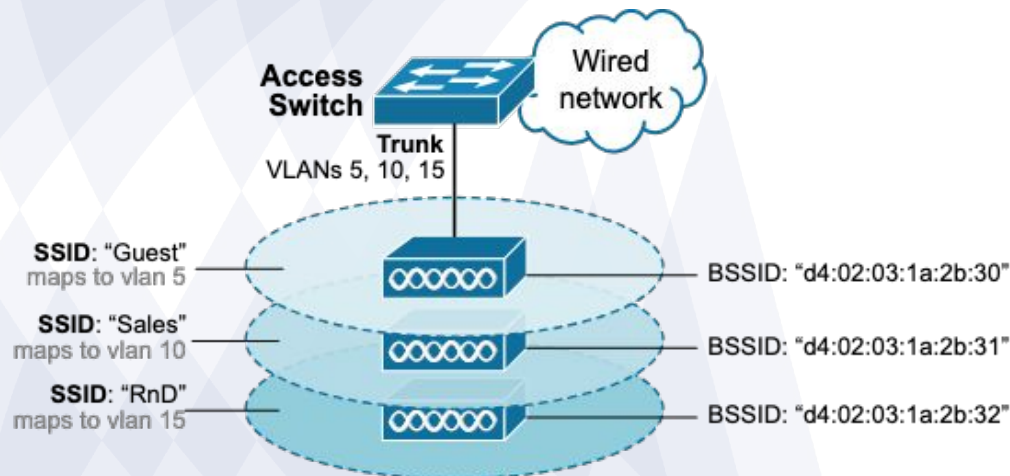


- From the Example (graphic 1), the clients PC 1, 2, and 3 connect to the wireless SSID "MyWiFi", while the clients PC 4, 5, and 6 connect via cables to the access switch. However, from an IP connectivity point of view, all six hosts are in Vlan 5 and have IP addresses from the same subnet, 10.5.1.0/24. They can ping each other and can reach every other address in Vlan 5.
- From the local router point of view, they are all in the same Vlan/Subnet (graphic 2)

WLAN Infrastructure: Autonomous AP Architecture

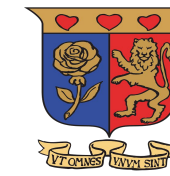


- In this context, the access point (AP) simply maps SSID "MyWiFi" to Vlan 5 and allows hosts to connect to that subnet wirelessly.
- Additionally, an autonomous AP can broadcast and support multiple logical wireless networks - multiple SSIDs. Each SSID can then be mapped to a different VLAN, allowing different groups of users to connect to different segments of the wired network. The diagram shows an AP that supports three SSIDs that map to different VLANs. If a client connects to SSID Guest, for example, it connects to an extension of the switched vlan 5.



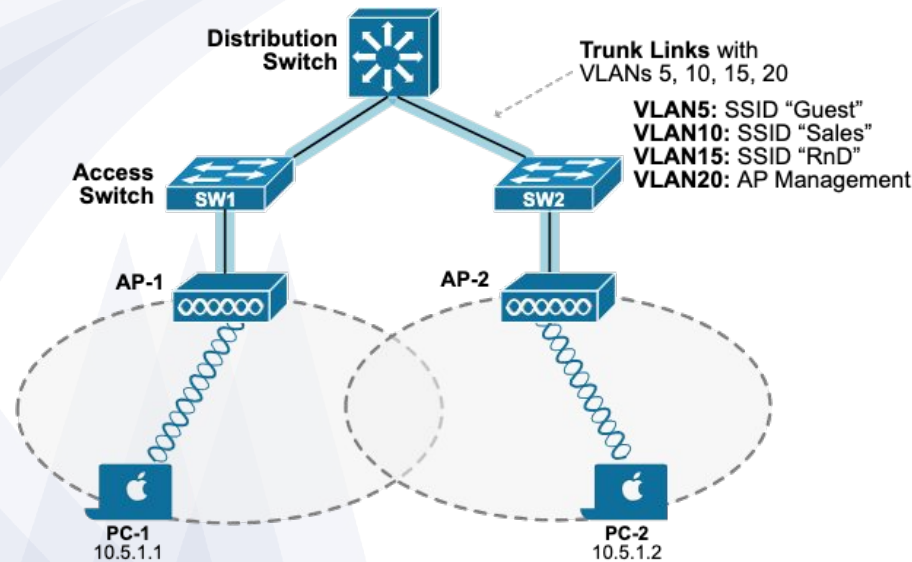
- Notice that for this setup to work, the VLANs must be trunked from the access switches to the AP. Pay attention to a common misunderstanding: having multiple SSIDs can create a false sense of scalability. Even if clients are spread across different SSIDs, they still rely on the same AP's hardware and compete for airtime on the same channel. This approach is used for logical segmentation of users, not for scale.

WLAN Infrastructure: Autonomous AP Architecture



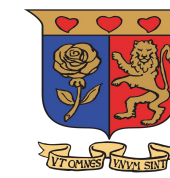
Strathmore
UNIVERSITY

- It is possible to build a sizeable wireless solution using access points working in autonomous mode. For example, the two APs connected to two different access switches, which are interconnected through a distribution switch. Both APs broadcast and support the same SSIDs as shown on the right side of the diagram..

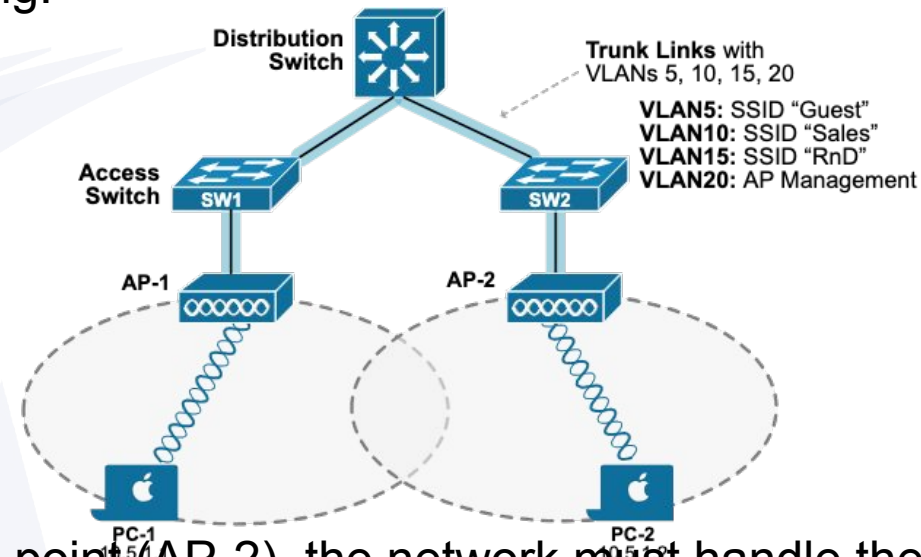


- Let's say that PC-1 connects to SSID "Guest" via AP-1, while PC-2 connects to the same SSID via AP-2. From an IP connectivity point of view, both hosts are connected to the same VLAN/Subnet. They can reach each other as though they are connected to two access switches in the same VLAN.

Autonomous AP Architecture: Client Roaming

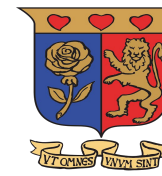


- One of the primary advantages of using wireless communication instead of wired one is that clients can move around while staying connected. This capability is called *roaming*. However, roaming imposes some network requirements for working.

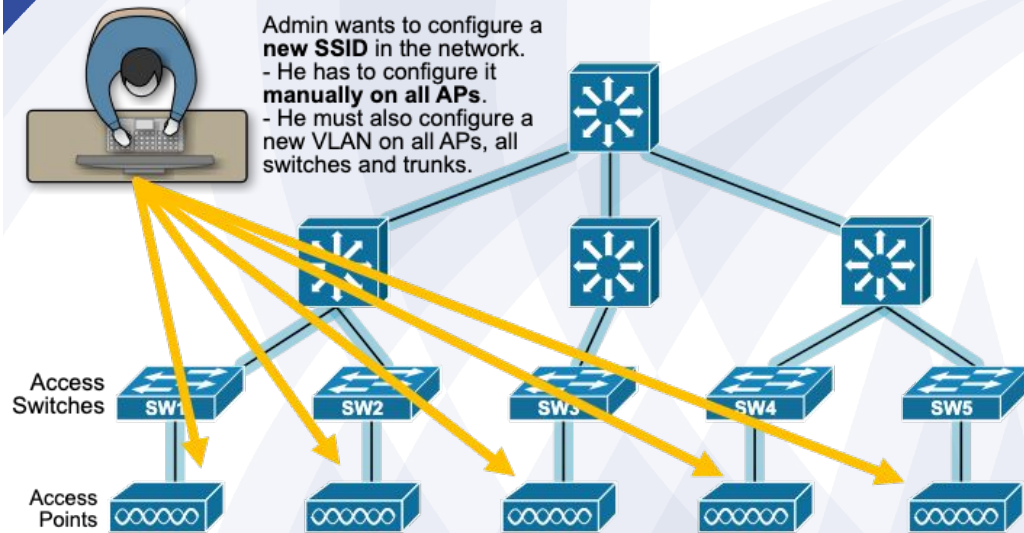


- When a client moves to a new access point (AP-2), the network must handle the transition. One important detail is the client's IP address. Before roaming, PC-1 is connected to AP-1 and gets an IP address from VLAN5. In the diagram above, the SSID "Guest" is mapped to VLAN 5, so the client receives an IP from the 10.5.1.0/24 subnet.
- When the client moves to AP-2, the SSID "Guest" is still mapped to VLAN 5 and the same subnet 10.5.1.0/24. Since the client remains on the same VLAN and subnet, it keeps the same IP address, and the transition happens very quickly, usually in less than 20ms. However, notice that this type of roaming requires that the VLAN mapped to the SSID must be trunked between the switches to ensure that the client stays on the same VLAN and can keep the same IP address.

Autonomous AP Architecture: Inefficiencies

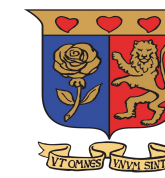


Strathmore
UNIVERSITY

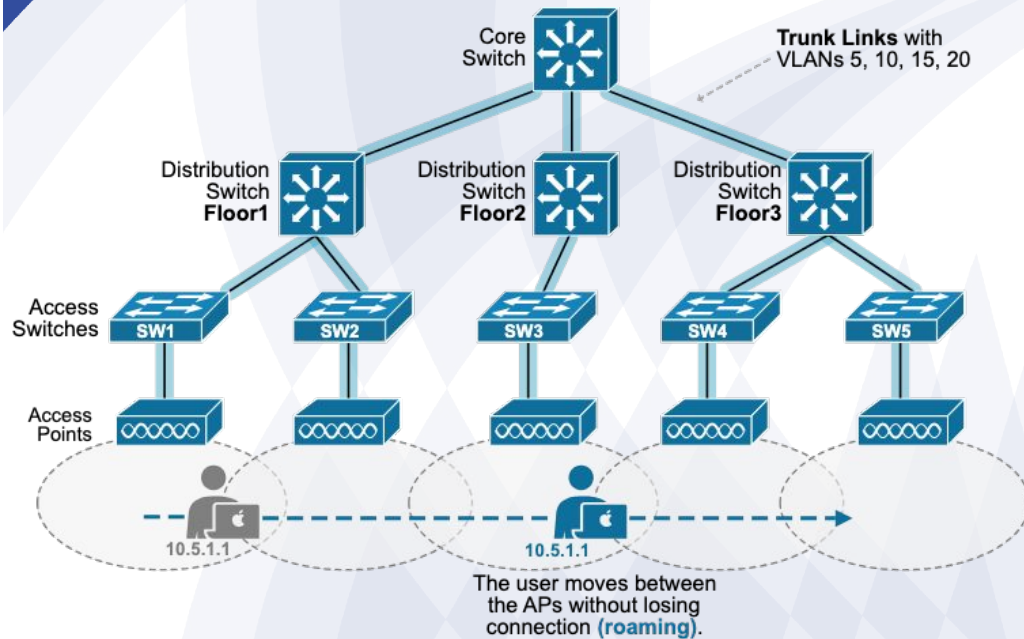


- **No centralized management:** Each autonomous AP is a standalone device. It has a management IP address for remote configuration. This allows you to configure parameters such as SSIDs, VLANs and wireless settings such as channel selection and transmit power. The management IP is usually separate from the data VLANs, so a dedicated management VLAN (VLAN 20 in our example above) is included in the trunk link to reach the AP. Since each AP operates independently, they must be configured one by one unless you use automation or a centralized management solution like the Cisco DNA Center, Ruckus smartzone.
- Imagine if an enterprise has hundreds of access points. Managing many autonomous APs can become a cumbersome and complex manual process. If the network administrator wants to configure a new SSID in the network - he has to configure it manually on all APs and then configure the corresponding VLAN (and its IP subnet) on all switches and trunks along the wired

Autonomous AP Architecture: Inefficiencies

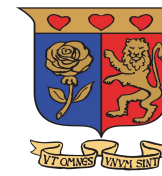


- **Roaming at scale:** The client mobility capability in the autonomous AP architecture requires layer 2 connectivity between all access points because a roaming client must remain on the same VLAN and subnet to keep the same IP address. The following diagram visualizes this requirement.



- Having the same VLAN stretched to all access points is a big scaling limitation. Most modern networks do not stretch layer 2 between different compartments of the network. For example, most modern designs use layer 3 routing between the distribution and the network's core.
- Additionally, what if you want to have the same SSID across multiple floors of an office building or even across multiple office buildings? You cannot simply stretch layer 2 across floors and buildings to allow the same SSID and provide roaming capability.

Autonomous AP Architecture: Inefficiencies

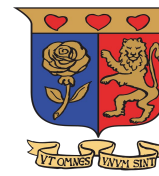


Strathmore
UNIVERSITY

Autonomous APs are used in a decentralized architecture, which has the following inefficiencies:

- Roaming requires stretching the VLANs and SSIDs on all APs. The VLANs must span the entire wired network, which is uncommon in modern network designs.
- If a network administrator wants to configure a new SSID, he has to create a new VLAN on all switches.
- The RF parameters of each AP are configured manually.
- No centralized node manages wireless traffic, such as policing, QoS, or intrusion detection. Everything is done manually.

Autonomous AP Architecture: Advantages

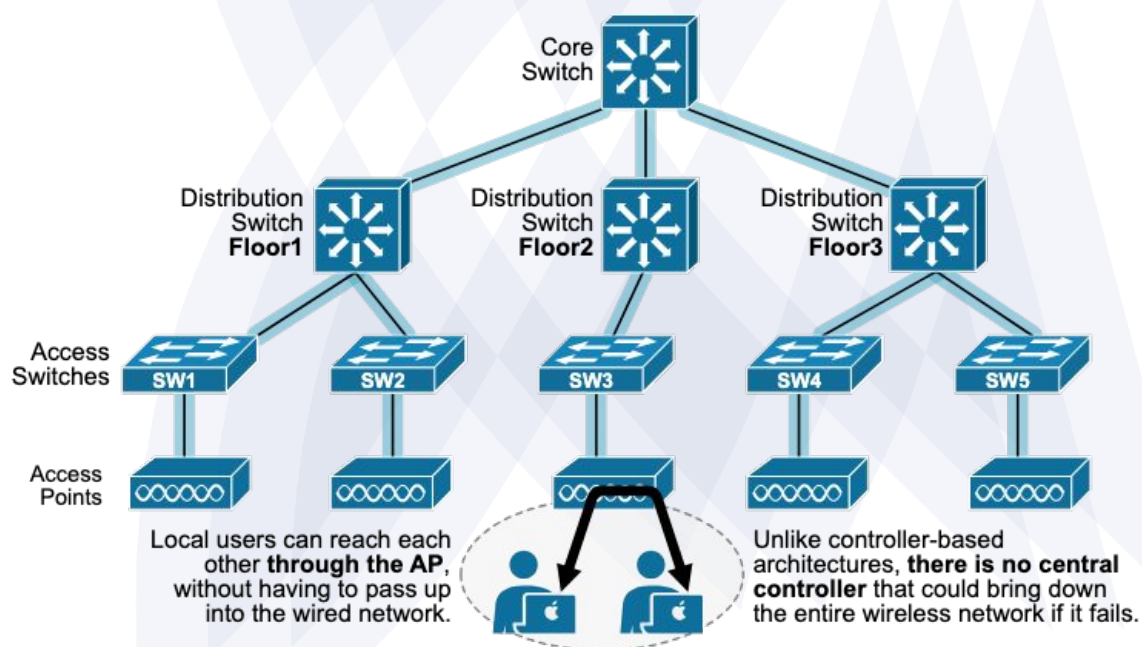


Strathmore
UNIVERSITY

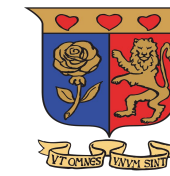
- One advantage of using autonomous APs is the direct and simple path between wireless and wired networks. For example, in the figure below, two wireless users connected to the same AP can communicate through it without sending data into the wired network. This makes sense since all wireless traffic must go through the AP in a Basic Service Set (BSS).

The autonomous AP architecture has several other advantages, such as:

- Independence** – Each AP operates on its own, meaning the failure of one AP does not impact others.
- Cost-Effective for Small Deployments** – It works well for small networks because there is no need for expensive wireless controllers and subscriptions for centralized management solutions.
- No Single Point of Failure** – Unlike controller-based architectures, there is no central controller that could bring down the entire wireless network if it fails.



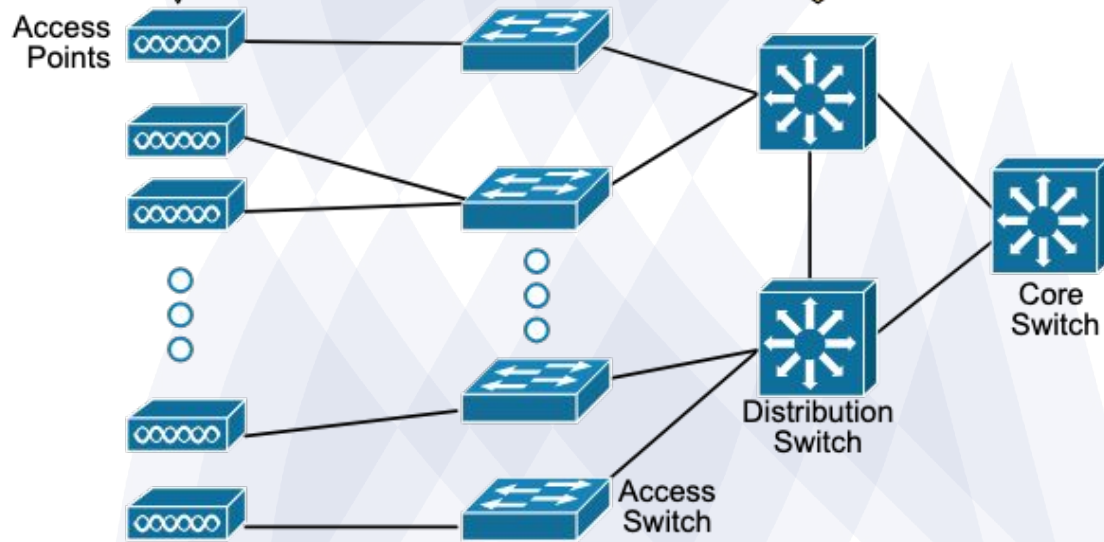
WLAN Infrastructure: Lightweight AP Architecture



Strathmore
UNIVERSITY

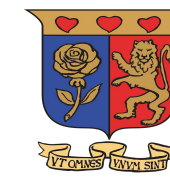
What if the organization has **hundreds of access points (APs)**?

And a **very large network** of access, core and distribution switches.

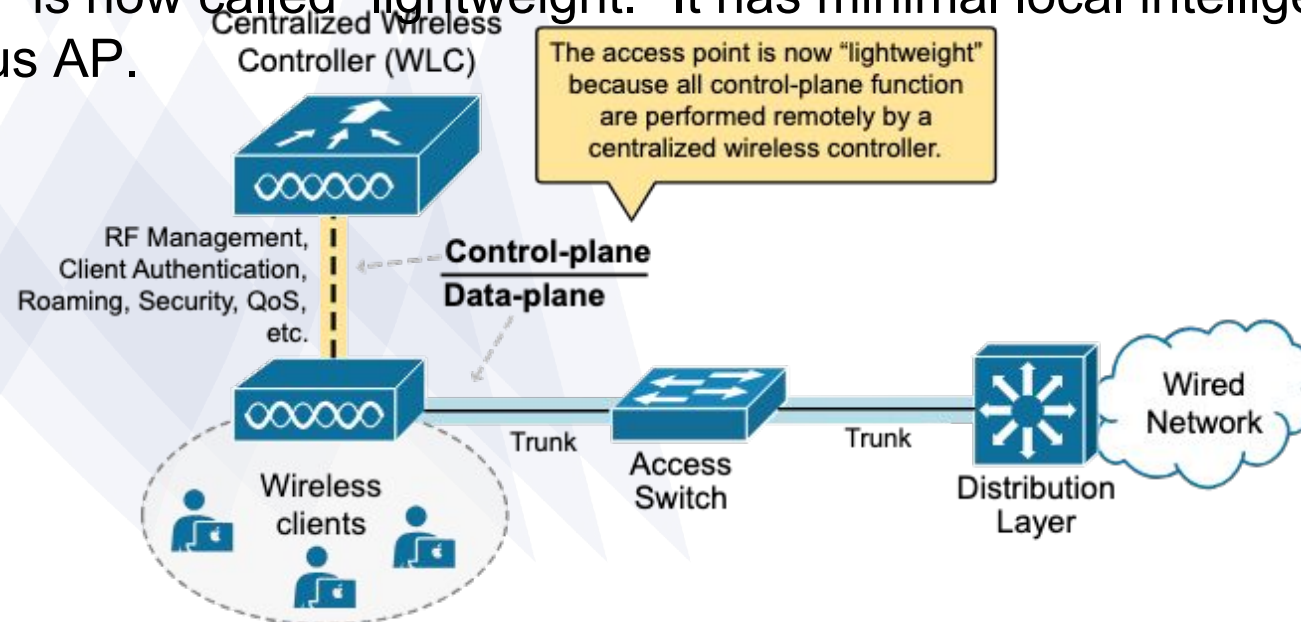


- An autonomous AP is managed as a standalone device; meaning every parameter (SSIDs, VLANs, IP addresses, Security mode, Encryption types, Channel settings, Transmit Power settings, QoS policies, ACLs, Failover settings, etc.) is configured manually directly on the device. This is very inefficient in a large network.
- For Example, Configuring a new SSID in the network will require hours and the process will be prone to errors.
- Additionally, it is difficult to collect information such as RF interference, channel usage, the number of devices, coverage etc.. It is also difficult to detect rogue APs and manage clients' authentication processes.
- There is need to move away from the decentralized configuration model (manually configuring AP per AP) and move to a centralized model where all access points are managed and operated from a

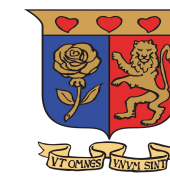
Lightweight AP Architecture: Separating Control and Data Planes



- To address the scaling limitations of the autonomous AP architecture, there is need to separate the control-plane functions of the autonomous APs into a centralized management controller, as shown.
- All control-plane functions are separated into a centralized wireless controller called WLC. It pushes the AP firmware and configuration and handles security policies, client authentication, RF management, and transmit power—making the access point fully dependent on it.
- The access point is responsible for Layer 1 and Layer 2 tasks, moving 802.11 frames in and out of the radio frequency environment and sending beacons and probe requests. Because of this, the AP is now called "lightweight." It has minimal local intelligence compared to an autonomous AP.

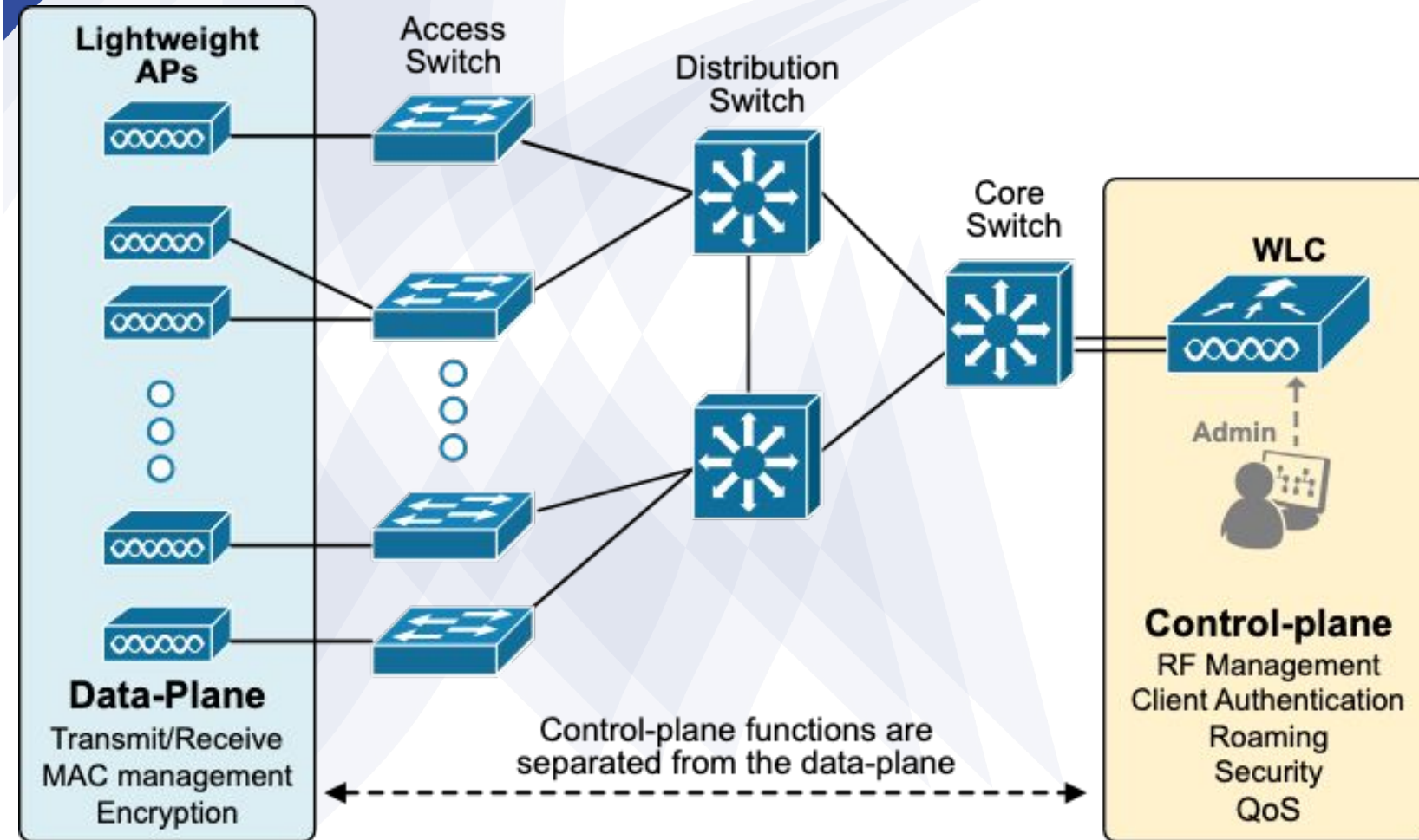


Lightweight AP Architecture: Separating Control and Data Planes

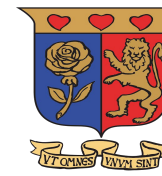


Strathmore
UNIVERSITY

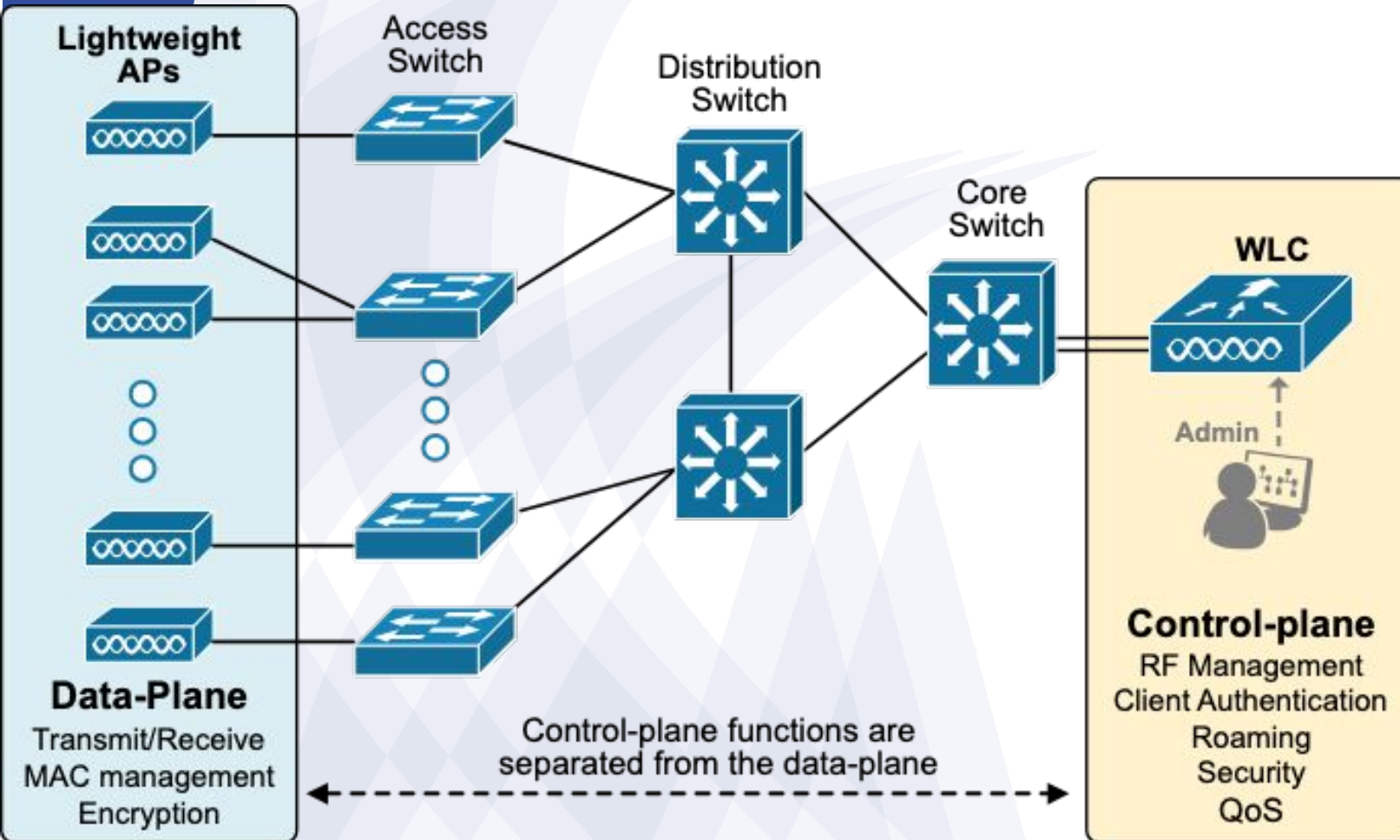
- This separation of functionalities allows for several significant scaling improvements.
- First, instead of configuring each AP individually, the WLC allows network administrators to manage all lightweight APs from a single node, as shown. This is a massive improvement in large-scale environments with hundreds of access points.
- If an admin wants to configure a new SSID, he doesn't have to configure hundreds of APs manually one by one.



Lightweight AP Architecture: Separating Control and Data Planes



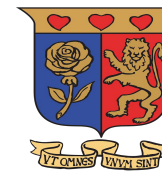
Strathmore
UNIVERSITY



- Another big improvement is the ease of deployment. Any new lightweight AP automatically connects to the WLC, downloads configurations, and starts operating without manual setup. This makes large-scale deployments much faster and more efficient.
- The controller also enforces security policies, including authentication, encryption, and rogue AP detection, ensuring consistent security across the entire wireless network.

This separation of functionalities between the "lightweight" access points (LAP) and wireless controller (WLC) is also known as a **Split-MAC architecture**.

Lightweight AP Architecture: Split MAC Architecture

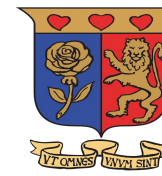


Strathmore
UNIVERSITY

- It is called the Split-MAC architecture because some of the Media Access Control (MAC) layer functionalities stay with the LAP while the controller performs others. The LAP handles direct interaction with clients and therefore real-time tasks related to wireless communication, such as beacons, probes, transmission, reception, fragmentation, and encryption of frames.
- Management tasks that do not require real-time handling of wireless frames are managed centrally. by the WLC, which controls multiple LAPs from a central location.

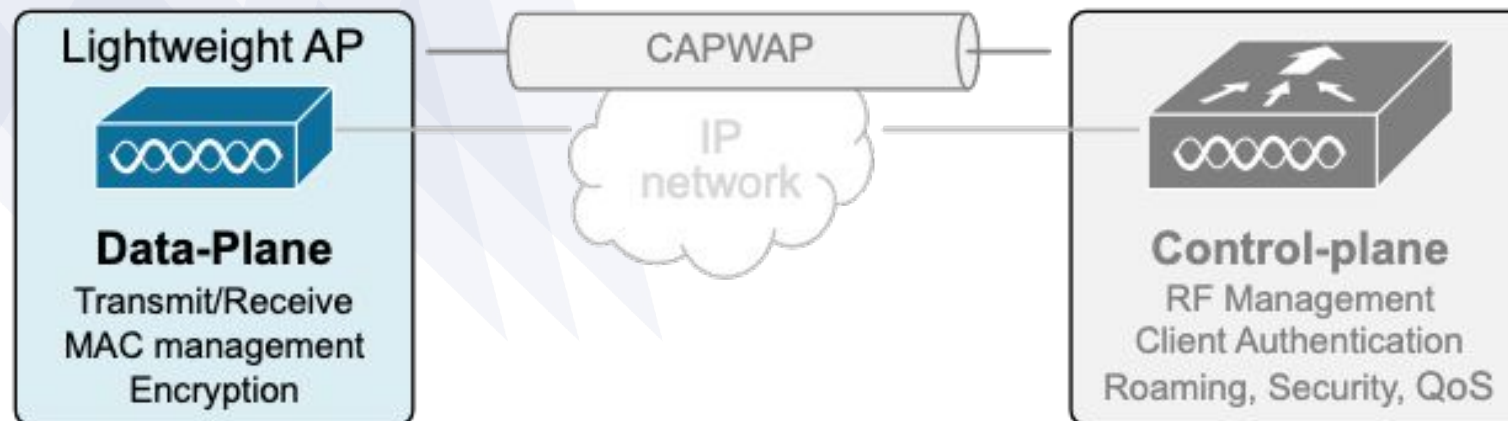
LAP MAC functions	WLC MAC functions
Sending wireless beacons and responding to probe requests	Handling client authentication
Frames acknowledgments and retransmissions	Managing client association and roaming
Managing frame fragmentation, queueing and prioritization	Translating wireless frames to other network protocols
Encrypting and decrypting wireless data	Terminating wireless traffic on a wired network

WLAN Infrastructure: Lightweight AP



- A lightweight AP is designed to work with a wireless controller (WLC) rather than operate independently. It handles real-time tasks like 802.11 frame transmission and encryption on the data plane while the control plane is separated into a remote centralized controller..
- Because they are "lightweight," LAPs rely entirely on a WLC for configuration and firmware management. They are deployed with a "zero-touch" approach, meaning they don't require individual setup.
- The AP and LAP are often the same hardware devices but run different firmware depending on their mode of operation. The autonomous AP comes with fully functional firmware that has all control capabilities while lightweight AP has minimal firmware that allows the AP to boot up, obtain an IP address, and communicate with a WLC to download the latest controller-based firmware.

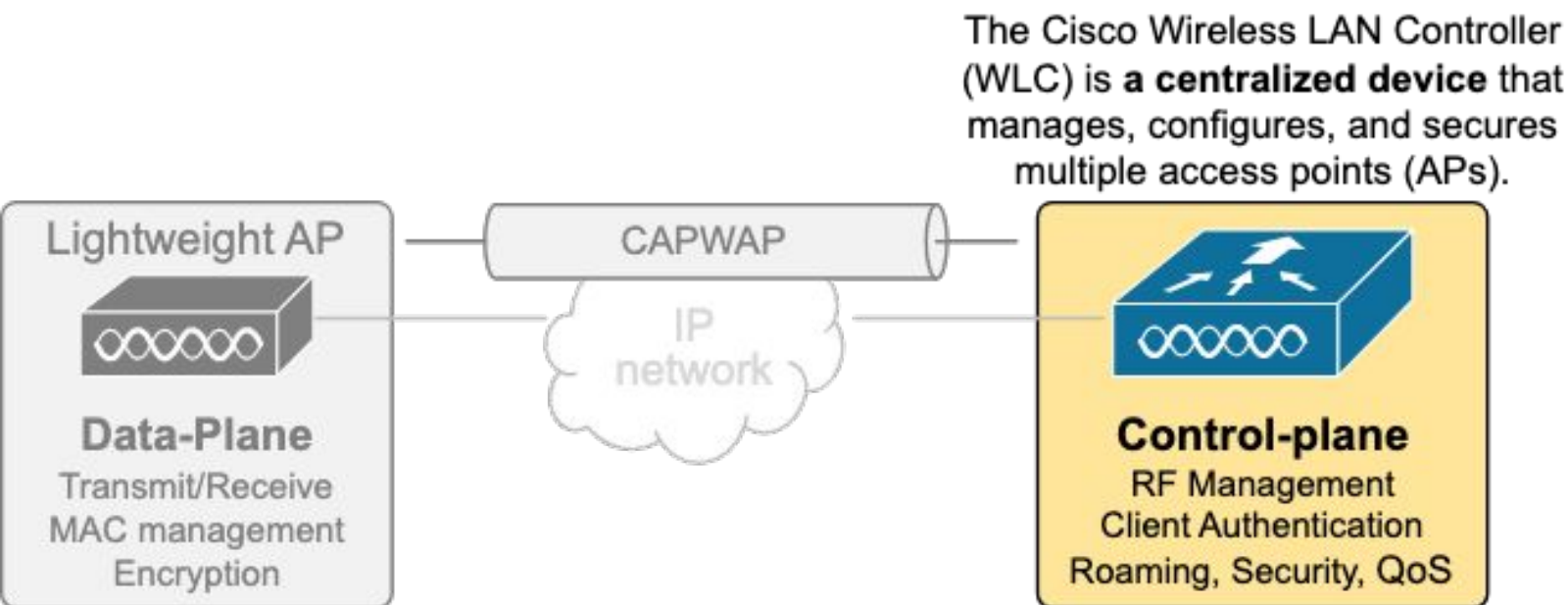
A LAP cannot operate on its own—it **has to join a WLC** to become fully functional.



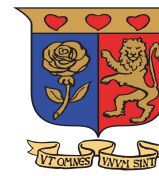
WLAN Infrastructure: Wireless LAN controller (WLC)

- The Wireless LAN controller (WLC) is a separate physical appliance that manages all lightweight access points (LAP) in a centralized manner. It is typically deployed in the organization's data center or regional hub. It provides a single-pane-of-glass interface for configuring and operating everything related to the organization's wireless network.
- This centralized wireless control makes the wireless network more scalable, efficient, and secure.
- The Wireless LAN Controller (WLC) solves many of the scaling problems seen in the traditional autonomous WLAN architecture including:

- **Automatic Channel Selection** – The WLC picks the best RF channel for each LAP based on nearby access points.
- **Transmit Power Control** – The controller adjusts each LAP's transmit power to provide the right coverage.
- **Self-Healing Coverage** – If a LAP fails, the controller increases the power of nearby LAPs to cover the gap.
- **Fast Client Roaming** – Clients can move between LAPs smoothly, even across Layer 2 and Layer 3 networks.
- **Load Balancing** – It distributes clients among LAPs to avoid overloading one access point.
- **RF Monitoring** – It listens to different channels to detect interference, noise, rogue APs, and other signals.
- **Security Management** – It verifies client authentication and ensures they get an IP from a trusted DHCP server.
- **Intrusion Protection** – It scans client traffic to detect and prevent security threats.

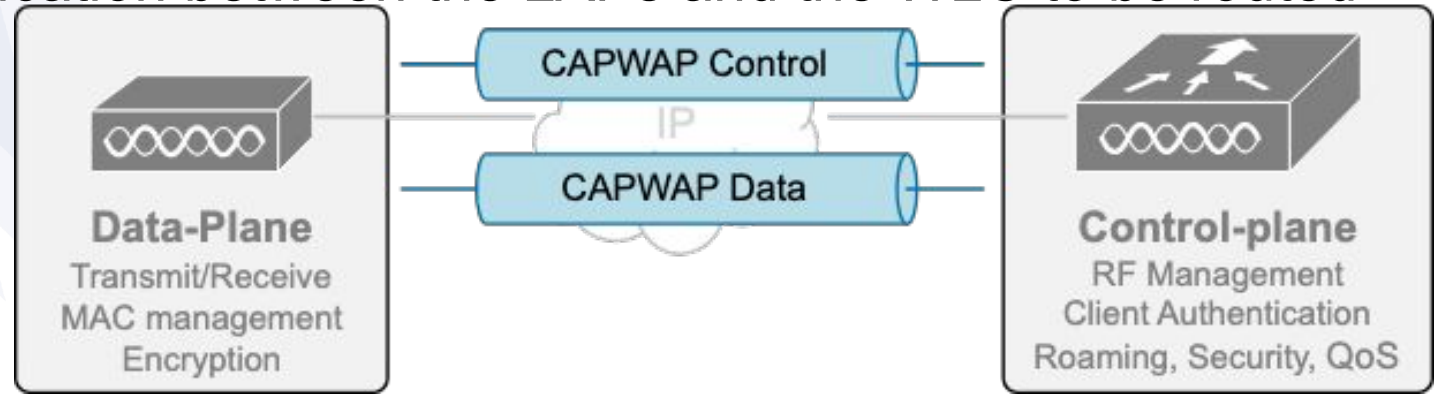


WLAN Infrastructure: Introducing CAPWAP

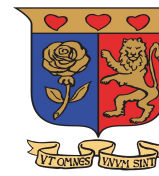


Strathmore
UNIVERSITY

- The centralized wireless architecture divides tasks between the Lightweight Access Point (LAP) and the Wireless LAN Controller (WLC). Each LAP must connect to a WLC to function and serve wireless clients. The WLC manages multiple LAPs across the network. *But how does a LAP connect to the controller? Do they need to be on the same network? Do they need to have layer 2 connectivity in between (all VLANs spanned to the controller)?*
- The wireless controller can be anywhere on the organization's network, even in the public cloud. The only requirement is that the LAPs must have IP connectivity to the WLC.
- To connect to the controller, the lightweight access points use a tunneling protocol called "***the Control and Provisioning of Wireless Access Points (CAPWAP)***" protocol." It makes this connection possible by encapsulating data into new IP/UDP packets, allowing the communication between the LAPs and the WLC to be routed across the network.
- Unlike autonomous APs, a lightweight AP sends the client's traffic to the WLC by default.
- The controller is responsible for bridging the client's traffic into the wired network, as shown.

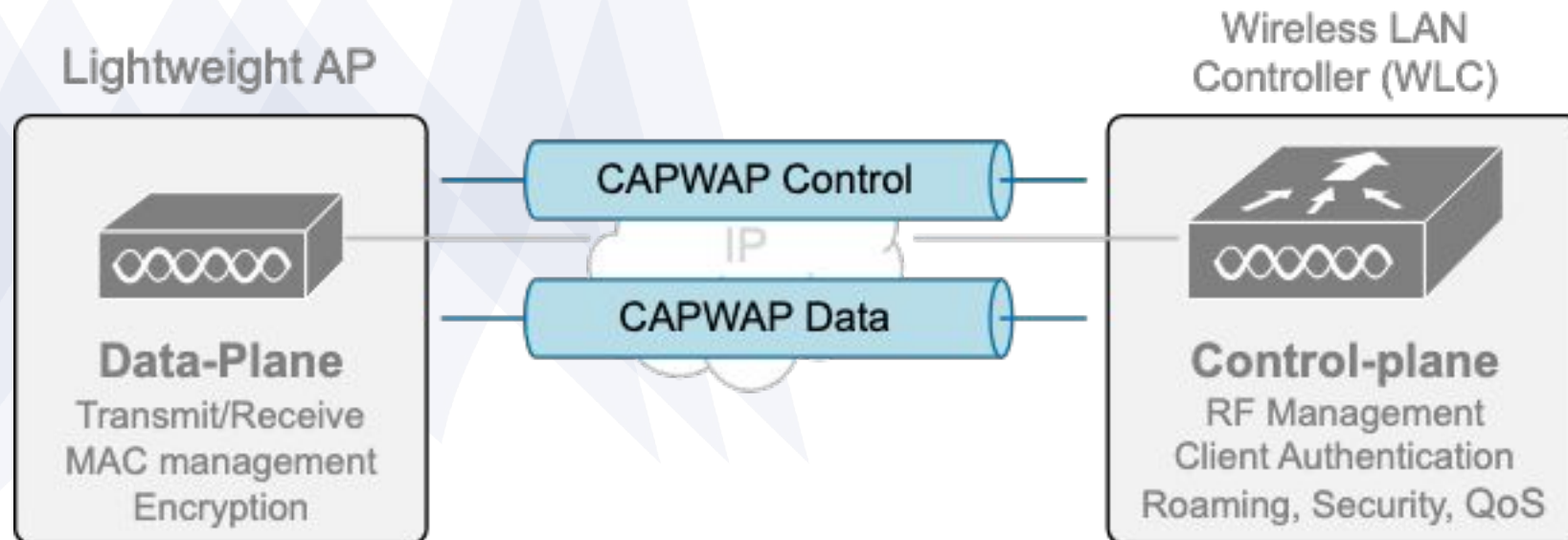


WLAN Infrastructure: Introducing CAPWAP

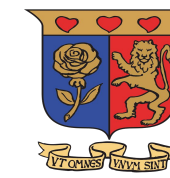


Strathmore
UNIVERSITY

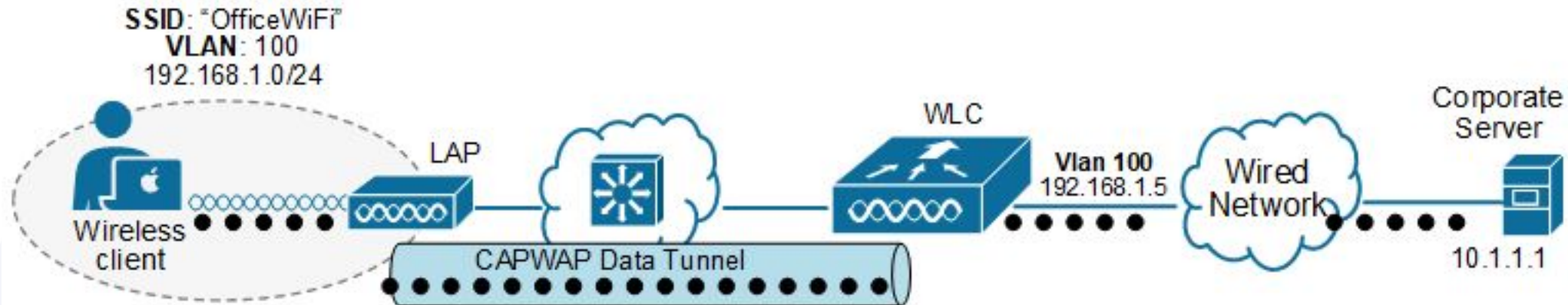
- Since control and data planes are separated, CAPWAP creates not one but two tunnels as follows:
 - Control Tunnel (UDP 5246): Used for configuring and managing the LAP. These messages are encrypted to ensure secure communication.
 - Data Tunnel (UDP 5247): Used to transmit client data. By default, data is not encrypted, but encryption can be enabled using Datagram Transport Layer Security (DTLS).
- Each LAP and WLC must authenticate using digital certificates. Every device comes with a preinstalled X.509 certificate. This ensures only authorized access points join the network and prevents the addition of rogue APs.



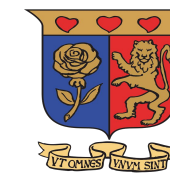
WLAN Infrastructure: Traffic Patterns with CAPWAP



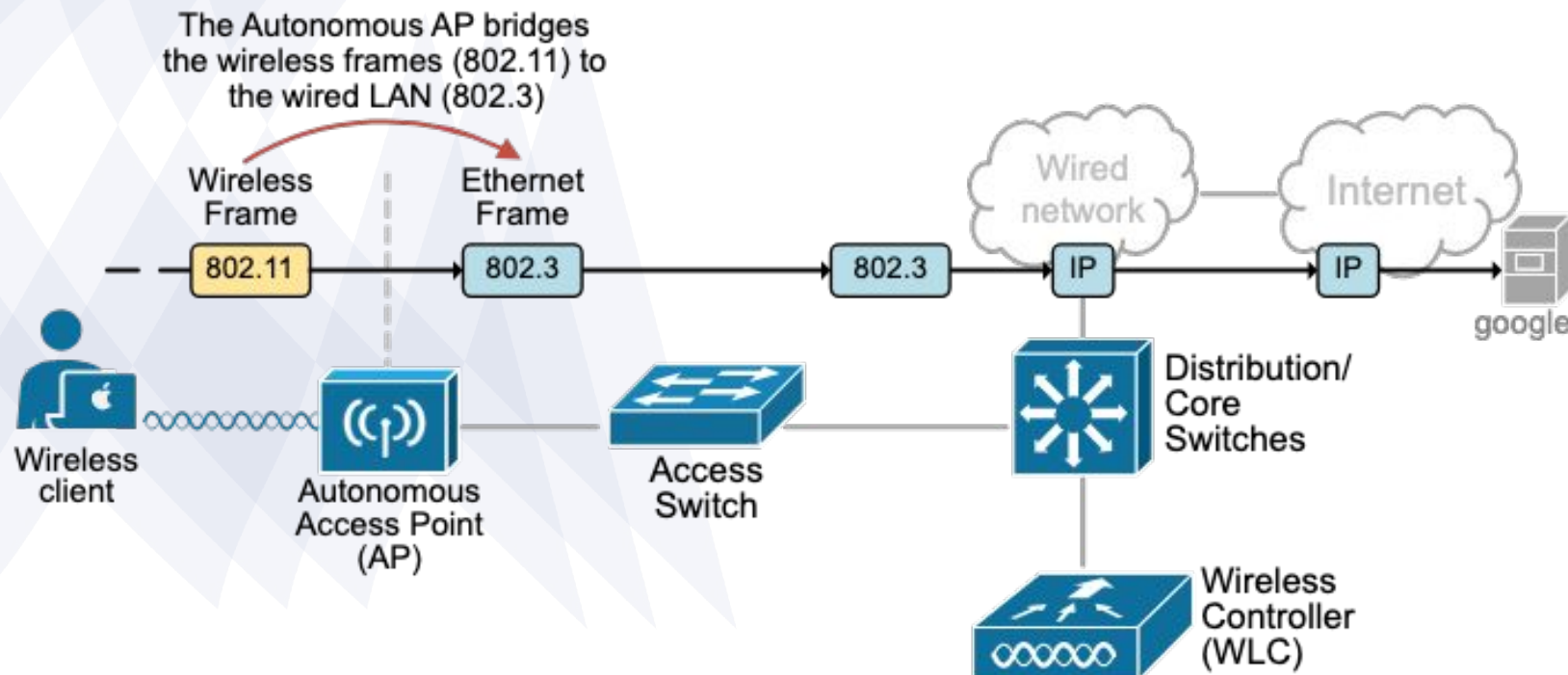
- Recall that autonomous access points directly bridge wireless clients's traffic to a wired VLAN at the upstream access switch. However, with the split-MAC architecture, CAPWAP tunnel changes the data traffic patterns of the access point as shown.
- Unlike autonomous APs, a lightweight AP sends the client's traffic to the WLC by default. The controller is responsible for bridging the client's traffic into the wired network, as shown.



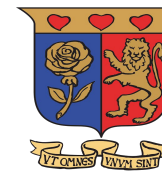
WLAN Infrastructure: Traffic Patterns with Autonomous AP



- Recall that an autonomous AP bridges wireless traffic from a wireless client to a switched network as shown.. When the client sends traffic destined to the Internet, the traffic goes to the AP. The AP directly forwards it to the wired network and out to the internet. The reverse traffic follows the same path from the Internet back to the wireless client.
- Notice that the frame format used inside the wireless network is 802.11, while the frames in the wired network use the Ethernet (802.3) format. They are different frame formats designed to serve different purposes based on the transport medium.



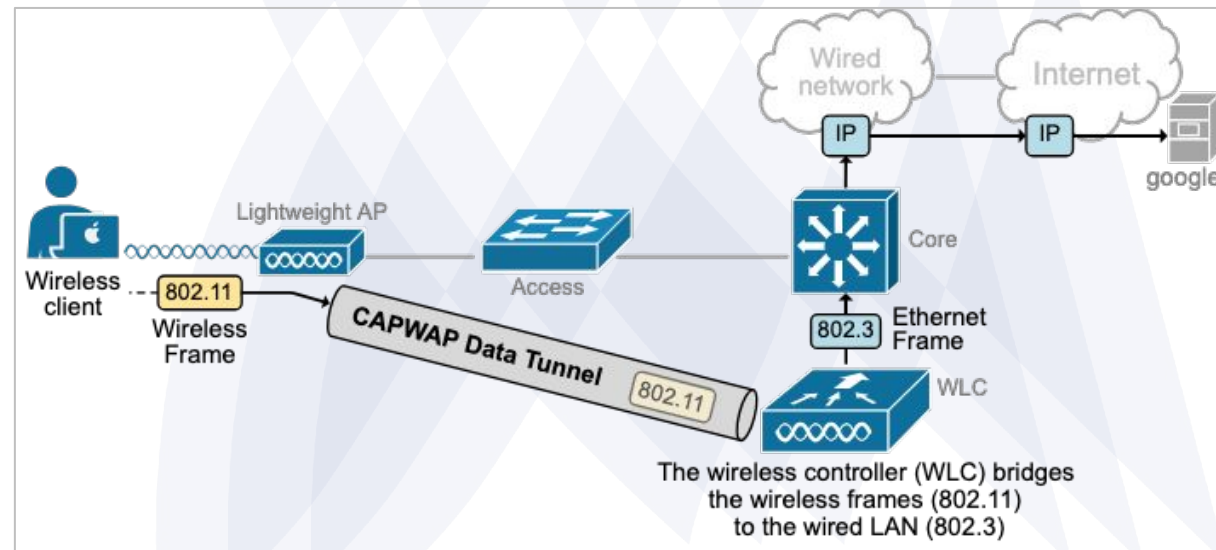
WLAN Infrastructure: Traffic pattern with CAPWAP



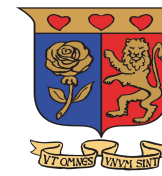
- In the same example, with the difference being that the AP is now "lightweight" and is managed by a centralized controller. With the split-MAC architecture, the LAP forwards the client's data through the CAPWAP data tunnel to the WLC, as shown.
- The lightweight AP encapsulates the client's traffic inside a CAPWAP data tunnel between the LAP and the WLC. This means adding a CAPWAP header (UDP 5247) with the source IP being the LAP's management IP address and the destination IP being the WLC management IP address.

The wireless controller removes the CAPWAP header and forwards the traffic into the wired network by converting it from 802.11 to 802.3 frames.

The original client's MAC address is not used on the wired network; instead, the WLC uses its own MAC address when forwarding traffic. However, the client's original source and destination IP addresses have not been changed.

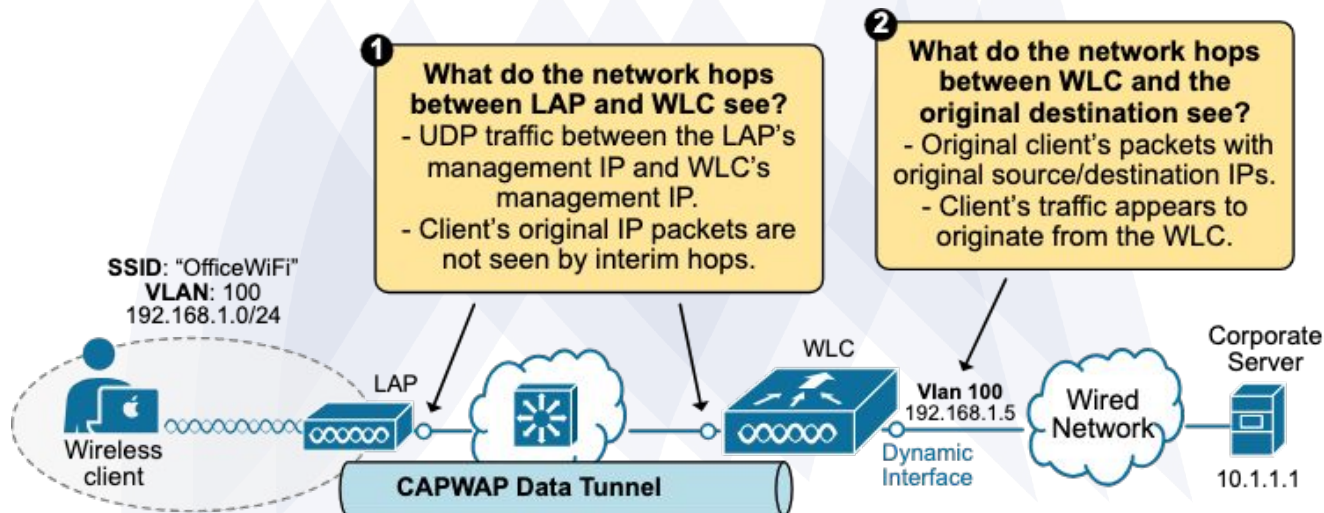


WLAN Infrastructure: Traffic pattern with CAPWAP



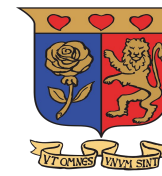
Strathmore
UNIVERSITY

- The result is that all wireless client traffic appears to originate from the WLC when it reaches the wired network. The controller removes the CAPWAP header and sends the original packet towards its destination. Therefore, the rest of the wired network sees the original packet coming from the wireless controller, as shown in the diagram below.
- The controller could be anywhere in the network. It doesn't have to be in the same subnet as the access points. It only needs IP connectivity to the LAPs' management IP addresses.

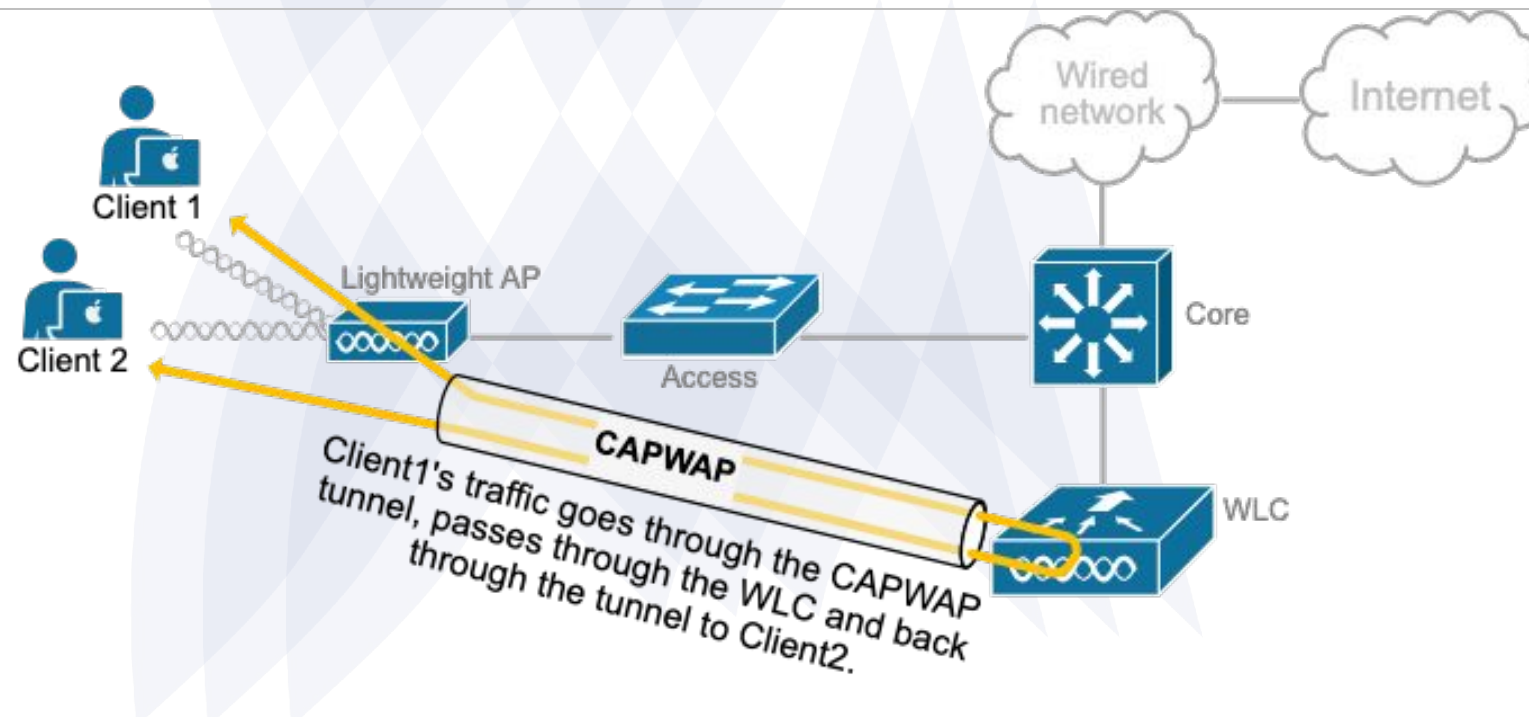


- The VLANs that match the available SSIDs don't have to be trunked to the lightweight access point. The LAP only needs one management IP address.

WLAN Infrastructure: Traffic Pattern between clients of the same AP

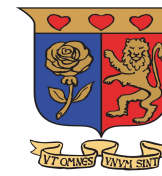


- Consider communication between two wireless clients that are connected to the same access point and associated with the same SSID. Although they appear to be local and very close together, the traffic between the two goes through the CAPWAP tunnel to the controller and then return back as shown.
- In a large campus network, this traffic pattern is not a big issue because the WLC is centrally located, and bandwidth is usually sufficient. But what happens if the network expands?



If LAPs are deployed at a remote site but the WLC remains at the main campus, all wireless traffic must travel through the CAPWAP tunnel between the remote site and the main campus—even if the traffic is meant to stay within the remote site. This can cause inefficiencies, especially if bandwidth is limited.

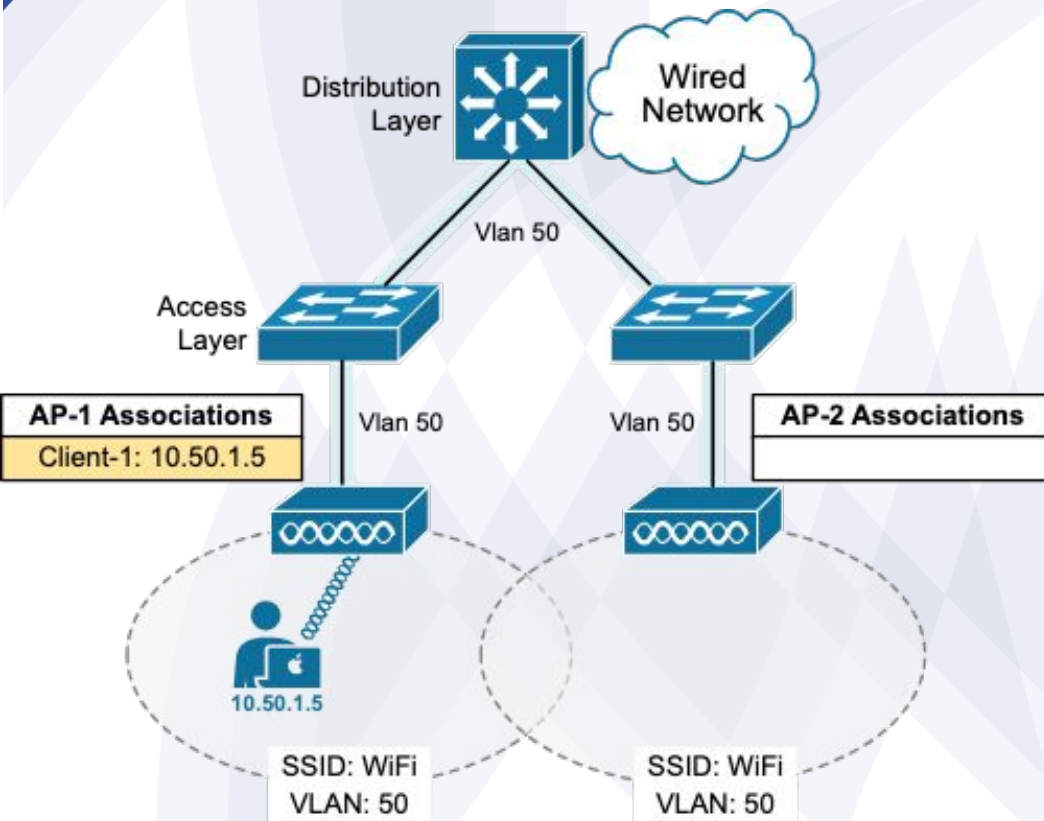
WLAN Infrastructure: Traffic Pattern between clients of the same AP



Strathmore
UNIVERSITY

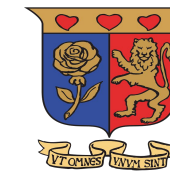
- FlexConnect mode can be used on remote LAPs. This mode allows traffic reaching the WLC to travel through the CAPWAP tunnel as usual. However, local traffic—such as communication between clients at the remote site—can be switched directly by the LAP without using the CAPWAP tunnel.
- Even if the link between the remote site and the main campus goes down, FlexConnect ensures that local wireless traffic continues to flow, keeping connectivity available within the remote site.
- If a LAP operates in FlexConnect mode, it can switch traffic locally instead of tunneling it to the WLC. This is useful at remote sites, where sending Internet-bound traffic through the CAPWAP tunnel to the central WLC would be inefficient. With FlexConnect, the AP can forward Internet traffic directly to the wired network at the local site.

Client Roaming: Roaming in the Autonomous AP architecture

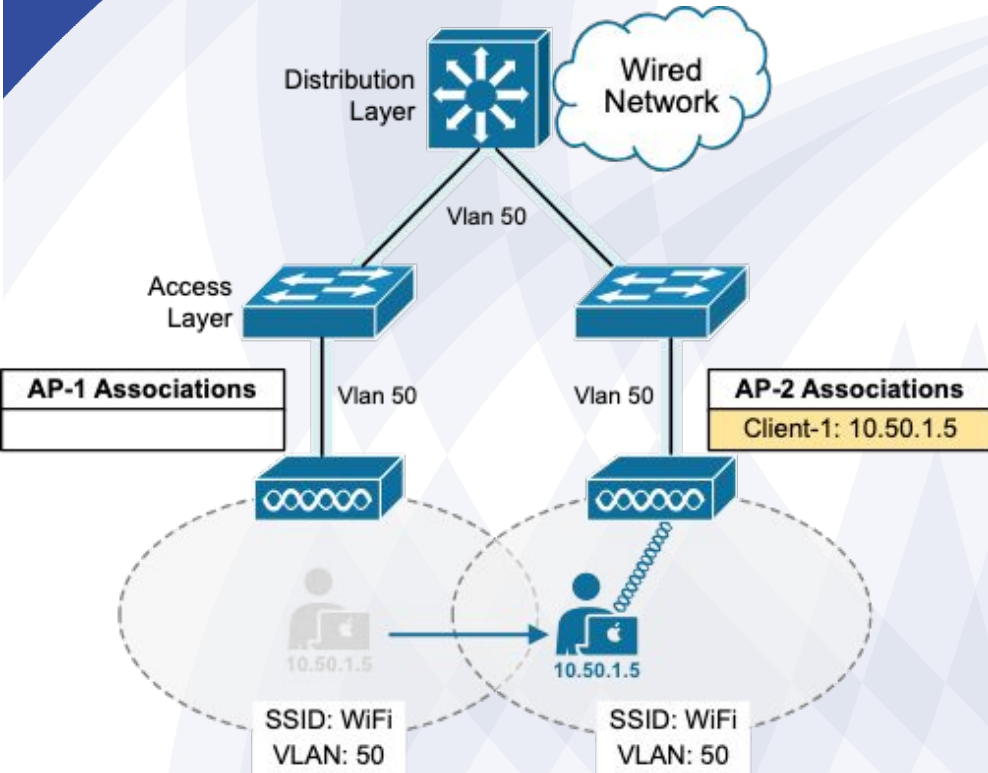


- When a wireless client moves around, it can switch from one BSS to another by roaming between access points (APs).
- The first fundamental part of the roaming process is that a client constantly checks its signal quality. If the signal weakens, the client looks for a stronger one from a neighbouring access point.
- The client passively listens for beacons and also actively sends probes to discover any available APs. If the client finds an AP with a stronger signal, it tries to re-associate with that access point.
- The diagram shows a simple topology with two access points and one wireless client. The client is associated with SSID "WiFi" via AP-1. The SSID maps to Vlan 50 upstream in the switched network. Since the access points operate in autonomous mode, each keeps a list of its connected clients. AP-1 currently has one client, while AP-2 has none.

Client Roaming: Roaming in the Autonomous AP architecture

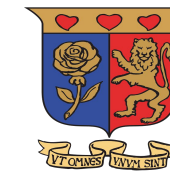


Strathmore
UNIVERSITY



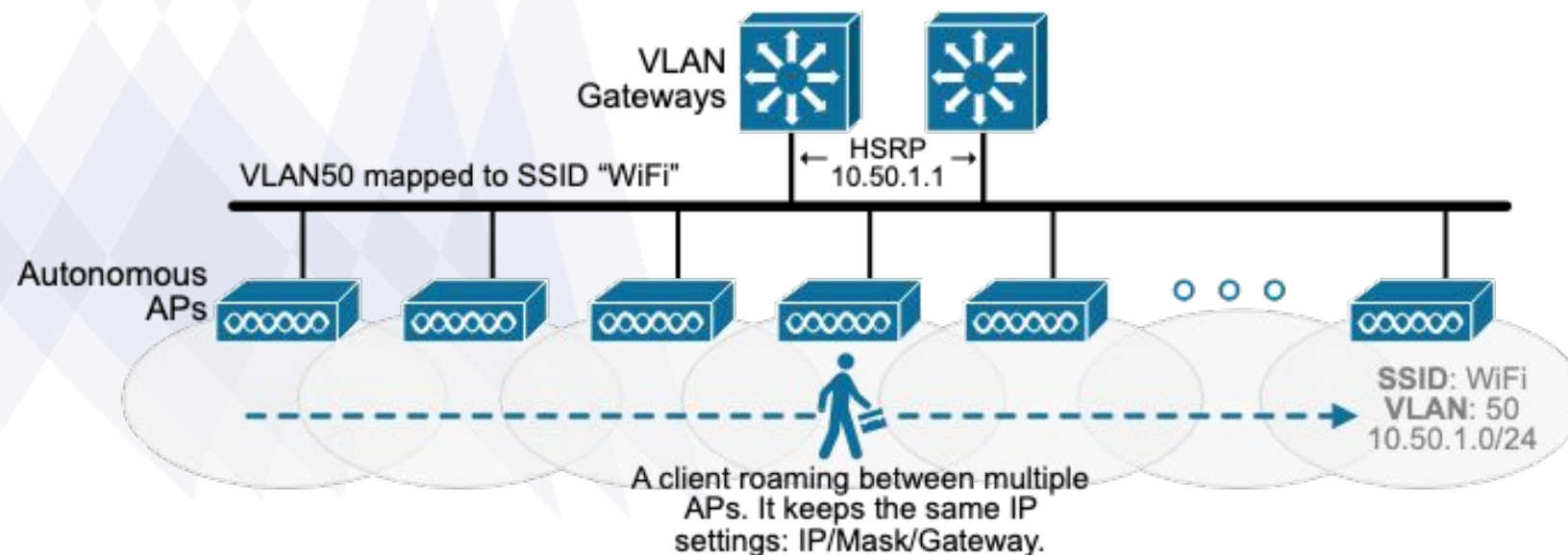
- When a client moves from AP-1 toward AP-2's coverage area. At some point, it detects a weaker signal from AP-1 and decides to switch to AP-2. The diagram shows the updated scenario. Both APs adjust their client lists to reflect the change. If AP-1 still has data for the client, it forwards it to AP-2 over the wired network.
- When moving around, the client decides when to roam based on its roaming algorithm. Some people wrongly assume that the access point decides to transfer the client to another AP. This is not the case. It is a decision of the mobile client - phone, tablet, laptop, etc.
- Additionally, keep in mind that roaming as a capability is vendor-agnostic. However, each client's decision about when to roam is vendor-specific. It is the secret sauce that makes the user experience during the roaming process and differentiates some vendors from others. Some vendors use the RSSI threshold as a trigger for roaming. Some compare the signal-to-noise (SNR) ratio. Some use packet loss and retransmissions to indicate that the client has to roam. Some use a combination of all, including roaming assistance from the AP.

Client Roaming: Roaming in the Autonomous AP architecture

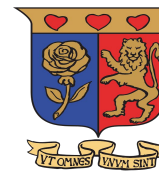


Strathmore
UNIVERSITY

- Roaming is not limited to two access points. In larger wireless networks, many autonomous APs can overlap to provide a continuous coverage area where clients can move around while staying connected. The following diagram shows how a client moves through multiple different APs.
- Notice that with the autonomous AP architecture, the VLAN that is mapped to the SSID that the client is associated with must be spanned to all APs to enable roaming. In practice, even though the clients connect to different access points, from a wired network's point of view, it stays connected to the same VLAN with the same IP settings.
- However, this design presents a scaling limitation because all large-scale networks do not span VLANs across different network compartments across many switches. Modern network design keeps the layer 2 domains small. That's why large-scale networks use the split-MAC wireless architecture.

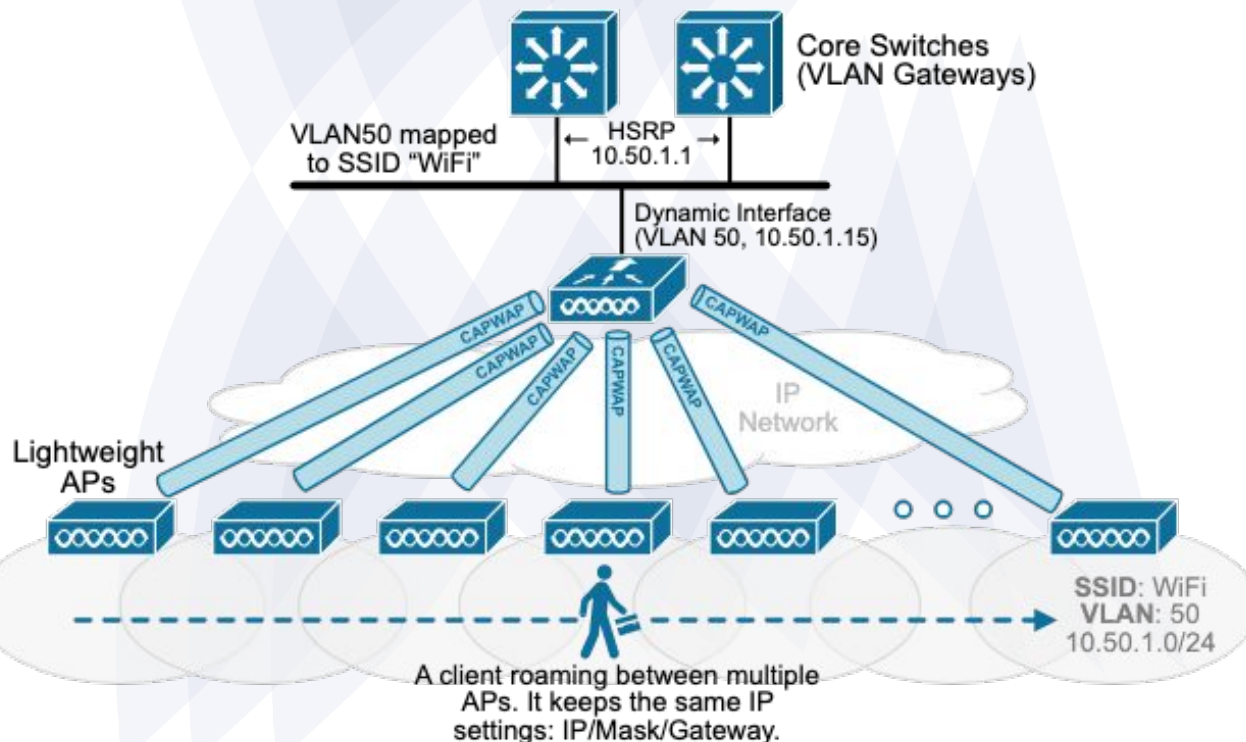


Client Roaming: Roaming in the Split-MAC architecture



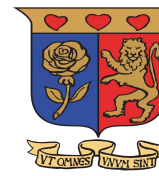
Strathmore
UNIVERSITY

- With the Split-MAC architecture, lightweight APs connect to a wireless controller using CAPWAP tunnels. Roaming works much like in autonomous APs—clients still need to reassociate with a new AP when moving. The key difference is that the controller manages roaming due to the split-MAC architecture.
- Let's first look at the overview high-level diagram below so that you can compare it with the autonomous architecture.



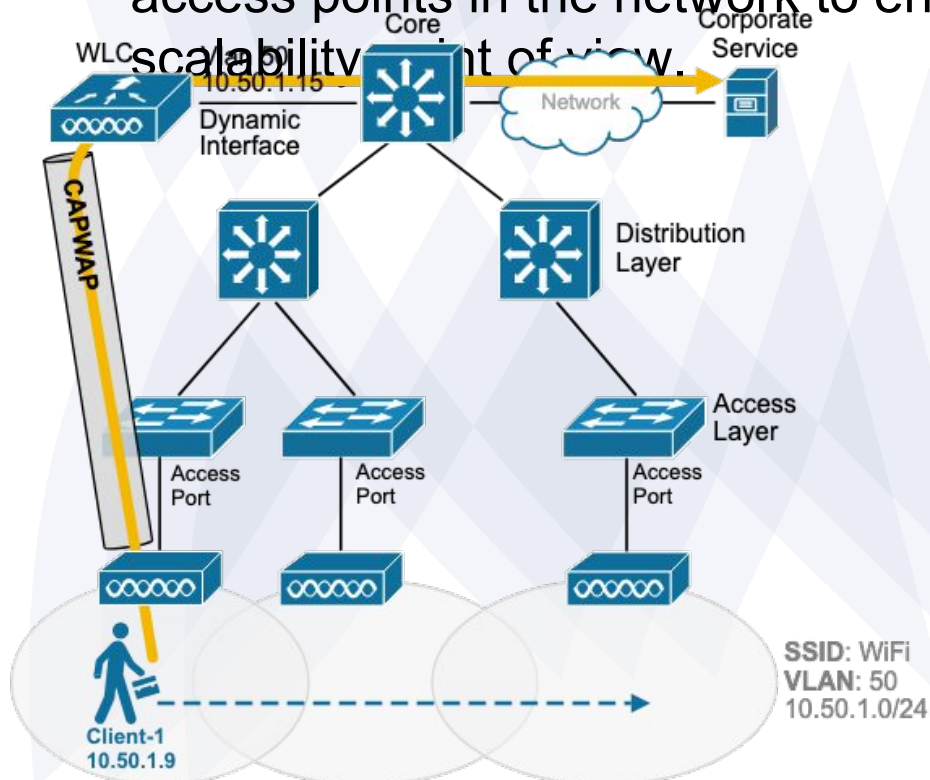
From the client's perspective, the roaming process is the same regardless of the wireless architecture. When the client detects that the Wi-Fi signals weaken, it initiates roaming based on its algorithm.

Client Roaming: Roaming in the Split-MAC architecture



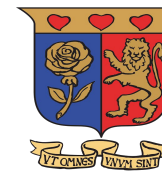
Strathmore
UNIVERSITY

- The diagram shows a simple setup with three lightweight APs linked to a single controller. Client-1 is connected to AP-1, which has a CAPWAP data tunnel to the wireless controller (WLC). The controller keeps a database with details about each client, including AP associations and SSID information.
- Notice the first big improvement, compared to the autonomous AP architecture - the VLAN (VLAN 50) that maps to the SSID ("WiFi") does not have to be spanned to all lightweight access points in the network to enable roaming. This is a significant improvement from a scalability point of view.



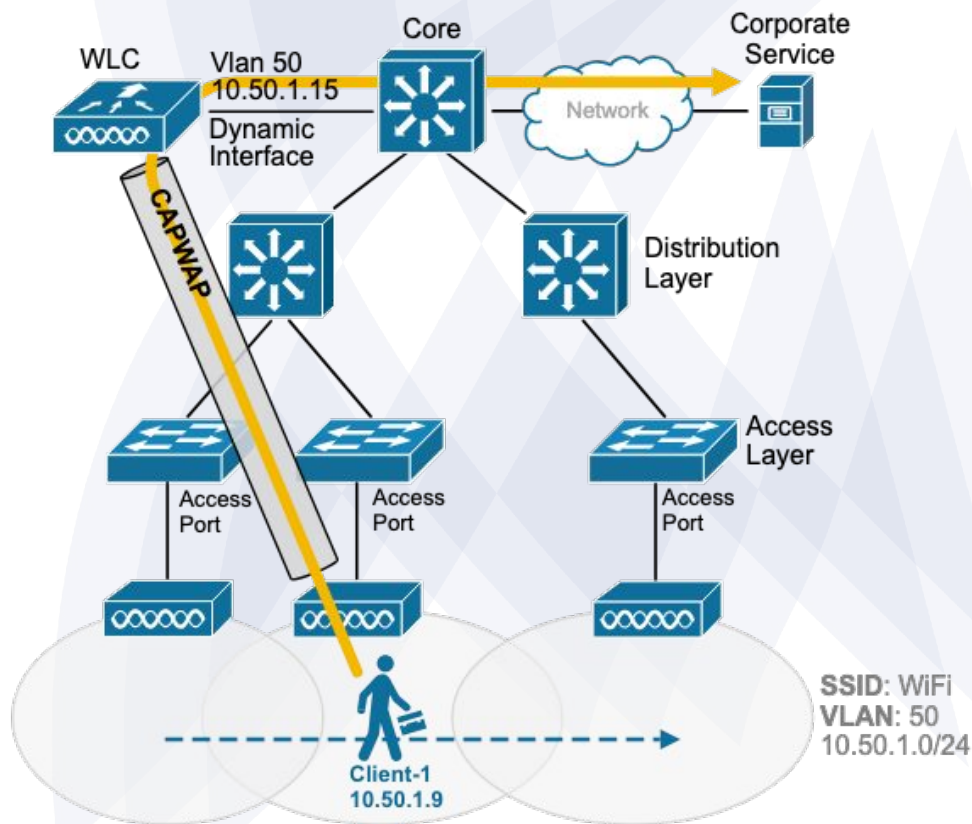
- Each lightweight AP is connected to its respective access switch in the management VLAN and only has one management IP address. VLANs that map to SSIDs do not need to be trunked to the access points.
- For example, VLAN 50 (10.50.1.0/24), which maps to SSID "WiFi," does not need to be trunked to the access points. The vlan is only present on the wireless controller, as shown.

Client Roaming: Roaming in the Split-MAC architecture



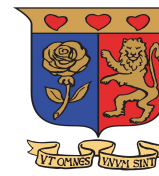
Strathmore
UNIVERSITY

- When Client-1 moves, it eventually roams to AP-2, as shown in the diagram below. The controller simply updates the client's association from AP-1 to AP-2. Since both APs are connected to the same controller via their CAPWAP tunnels, the entire process happens within the same WLC. This is called intra-controller roaming.

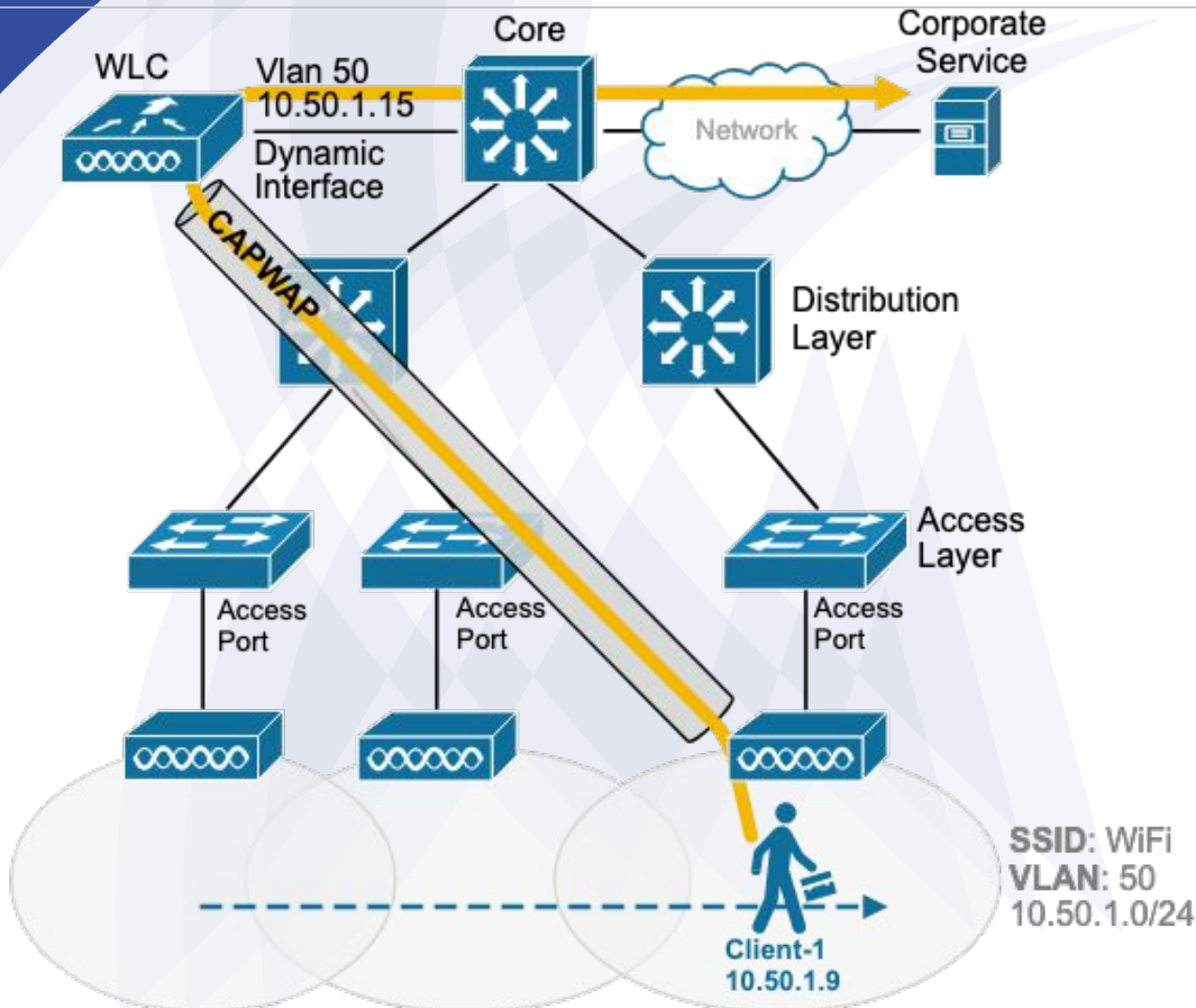


From a network point of view, nothing changes. Client-1 appears to be connected via the wireless controller's dynamic interface in VLAN 50. The client keeps the same IP settings: IP address, Mask, Gateway.

Client Roaming: Roaming in the Split-MAC architecture

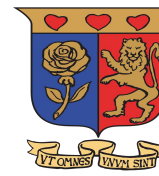


Strathmore
UNIVERSITY



- When Client-1 eventually roams to AP-3, as shown in the diagram, the controller simply updates the client's association from AP-2 to AP-3. Again, the entire roaming process happens within the same WLC.
- Since all APs are connected to the same controller, roaming is quick and seamless. The controller only needs to update its database to point to the correct CAPWAP tunnel, as shown in the diagrams.
- This process takes less than 10 ms. From the client's perspective, nothing changes—it just moves between APs based on signal strength without knowing about the controller or CAPWAP tunnels.

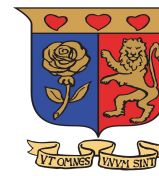
Client Roaming: Roaming in the Split-MAC architecture



Strathmore
UNIVERSITY

- Fast roaming is crucial for time-sensitive applications like wireless calling. A small delay during roaming can cause call interruptions or poor audio quality. Keeping this delay minimal ensures a smooth experience. During roaming, the controller may require 802.1X authentication, which involves communication with a RADIUS server and key exchanges.
- To speed up roaming, controllers support three key features:
 - Centralized Key Management (e.g. Cisco's CCKM) – One controller stores client keys and shares them with other controllers during roaming. This requires Compatibility Extensions (e.g. CCX).
 - Key Caching – Clients keep a list of keys from previous AP connections and present them when roaming. The new AP must be on this list, which can hold up to eight entries.
 - 802.11r – This fast-roaming standard allows clients to store part of the authentication key and reuse it when roaming. It also preserves QoS settings.
- For these fast-roaming methods to work, the client device must support them. It needs the right driver or supplicant software to handle fast authentication and key caching.

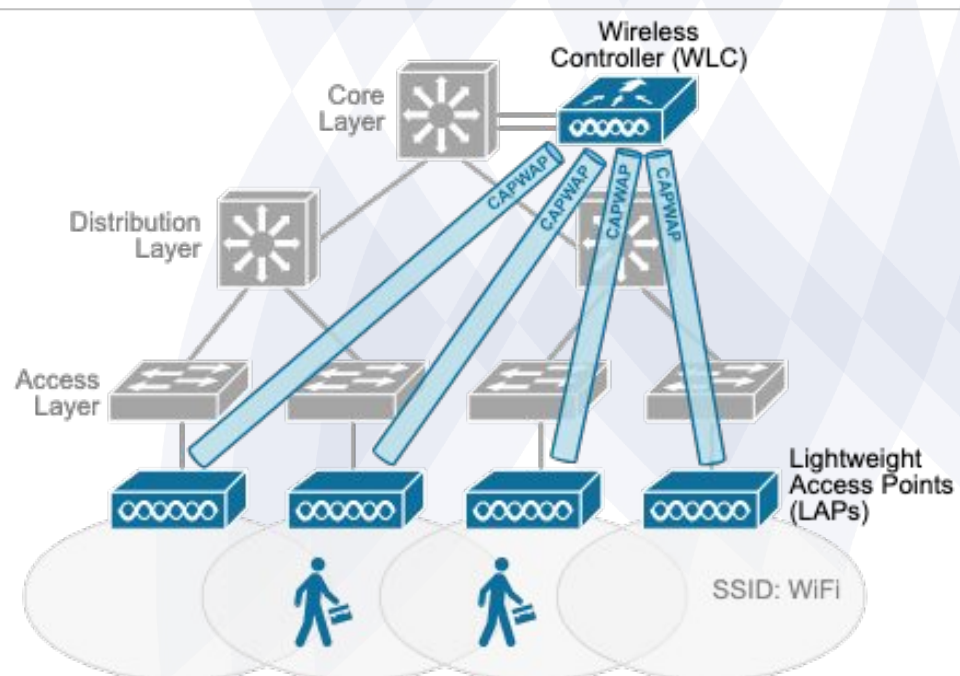
Wireless Controller placement: WLC at the Core Layer



- The placement of the wireless controller affects the performance and scale of the Wi-Fi network. There are four common placement points for connecting the WLC

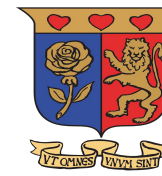
WLC at the Core Layer

- The most common design is to connect the wireless controller to the Core layer of the organization's data center or regional hub, as shown. This design makes a lot of sense in large-scale deployments with hundreds or thousands of access points. In such cases, scalability and available bandwidth at the controller's connection point become important.
- Since each lightweight access point sends the clients' traffic through the CAPWAP data tunnel to the controller, the traffic volume at the WLC can become very significant.



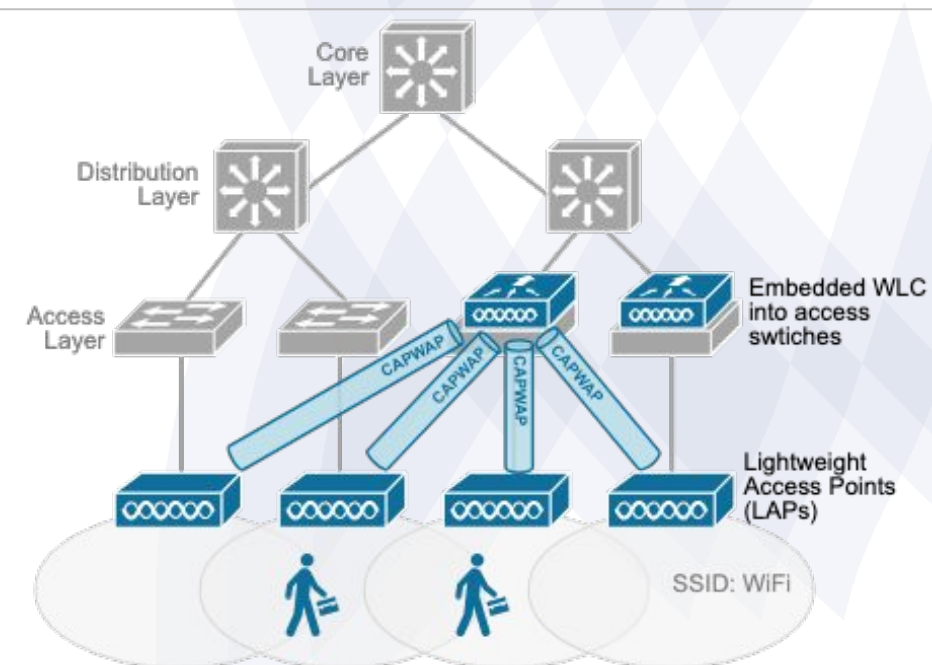
- For example, if the organization has hundreds of APs, the client's traffic reaching the controller may be in the range of 10+Gbps.
- Since the access and distribution layer simply might not have 40/100Gbps ports to connect the controller, the only appropriate place is the Core of the network.
- The main advantage of this design is that the WLC is connected to the high-speed backbone and can handle a lot of traffic.
- The main disadvantage is that the client's traffic gets further away from the client itself, and the traffic must transit the entire backbone network and back down in case of client-to-client traffic.

Wireless Controller Placement: WLC embedded at the Access Layer



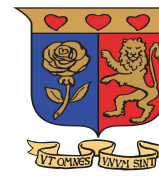
Strathmore
UNIVERSITY

- If most of the organization's wireless traffic is between clients connected via different access points, it makes sense to locate the wireless controller further down the network topology and closer to the clients themselves. As such a design where the WLC is built into an access switch is considered.
- With this design, the controller sits closer to the APs, shortening the CAPWAP tunnel. The CAPWAP encapsulation is only used between the AP and the upstream access switch, which is simply the LAN cable's length.

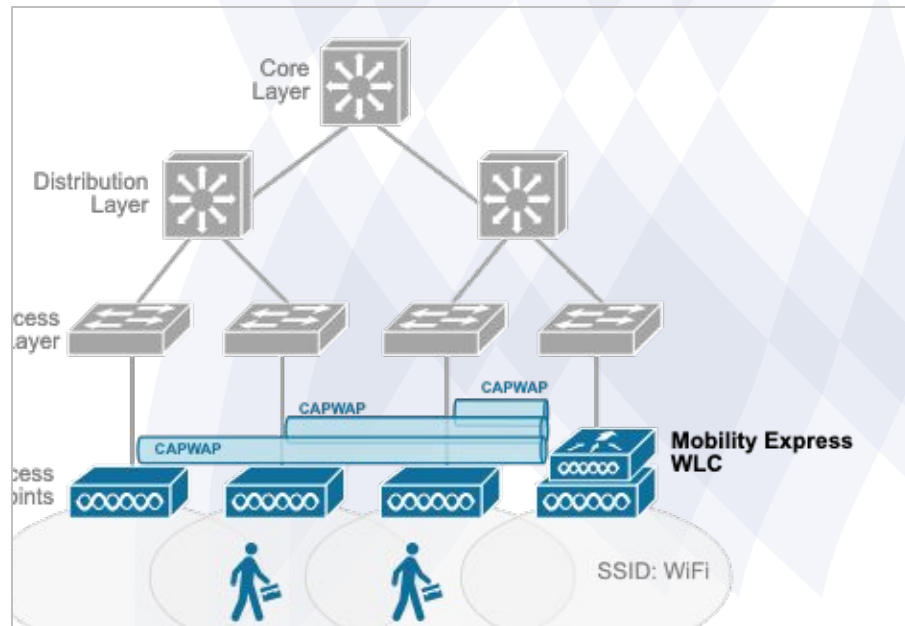


- This design can be cost-effective because the same switch handles both wired and wireless traffic, and the organization doesn't need to buy an expensive WLC appliance.
- It works well for branch offices and campuses. Since the WLC is local, APs do not rely on a distant WLC over a WAN connection. This improves reliability. Traffic between wireless users stays within the local network, while traffic to external resources is CAPWAP decapsulated directly at the access layer.
- The main advantage of this design is that it can be very cost-efficient for small to medium deployments of up to 200 access points.
- The main disadvantage is scalability, so it is not ideal for large enterprise networks.

Wireless Controller Placement: One Master AP as WLC

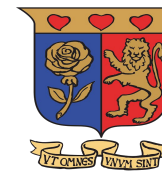


- The third design option involves a master-slave configuration of APs. It is exemplified in Cisco Mobility Express. It is a lightweight wireless solution designed for small—to medium-sized organizations that don't have the staff and resources to manage the complexity of a dedicated controller. Instead of requiring a separate controller (WLC), one of the access points takes on the role of the controller(master) while also functioning as a regular AP. This eliminates the need for additional hardware and simplifies network deployment, as shown.
- Once configured as master, the AP manages the rest of the APs in the network. Even if it fails, another AP can take over the role without disrupting the wifi network.



- This supports most of the advanced enterprise features that dedicated WLC does, such as fast roaming, VLAN segmentation, and RF management.
- However, it is limited in scalability compared to full WLC-based solutions, making it ideal for deployments with up to a hundred APs.
- The main advantage of Mobility Express is that it offers a cost-effective alternative to traditional controller-based solutions. It provides the flexibility of an enterprise-grade wireless network while maintaining the simplicity of an autonomous AP deployment.

Wireless Controller Placement: WLC in the Cloud

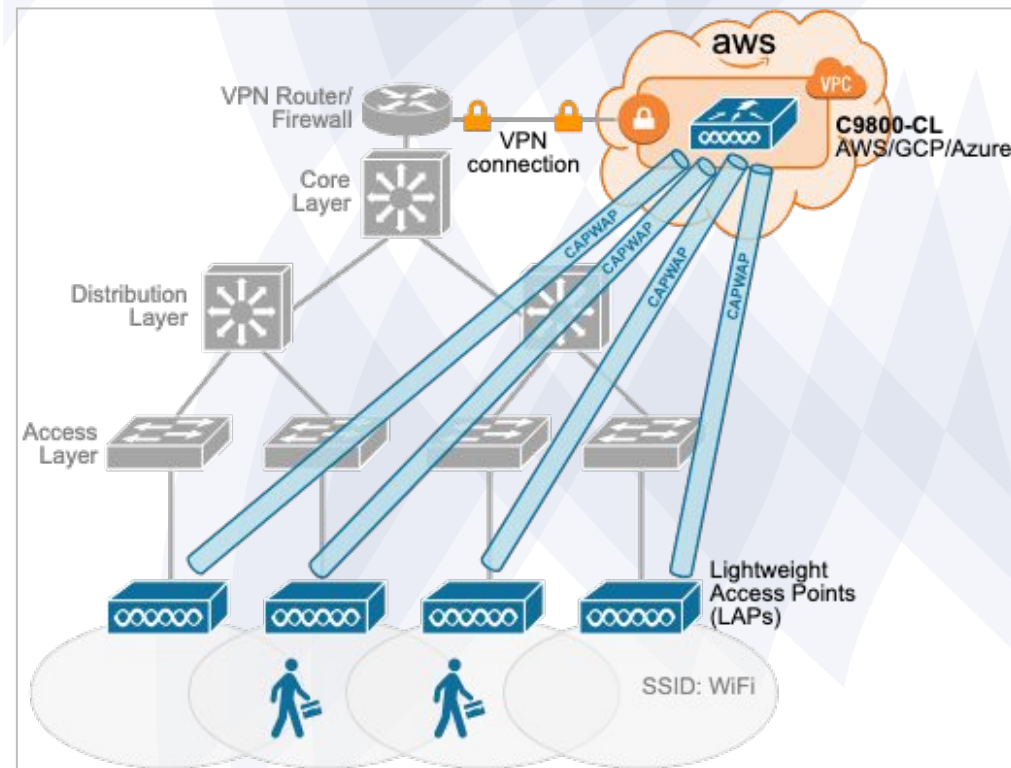


Strathmore
UNIVERSITY

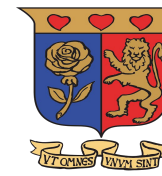
- Most WLC placement designs focus on placement of a wireless controller on-prem. It is however possible to have cloud-based designs. A virtual image can be deployed in a private or public cloud e.g. the Catalyst wireless controller whose latest version (Cisco Catalyst 9800-CL) is a cloud-ready virtual WLC running on VMware ESXi, KVM, or public cloud providers like AWS and Azure.
- Additionally, there could be a fully cloud-managed Wi-Fi solution (such as Cisco Meraki) that provides centralized control, security, and analytics for enterprise networks without requiring on-premises controllers.

Virtual Wireless Controller (vWLC)

- The virtual wireless controller (vWLC) offers several advantages over traditional hardware-based controllers. It provides flexibility by running in public cloud environments, allowing organizations to scale wireless networks without the need for physical WLC hardware.
- It also reduces capital expenditure (CapEx) since no dedicated hardware is required.
- Additionally, it supports high availability across multiple data centers, enabling fast failover and disaster recovery.
- The diagram shows a high-level of a cloud-based WLC design in Amazon AWS.

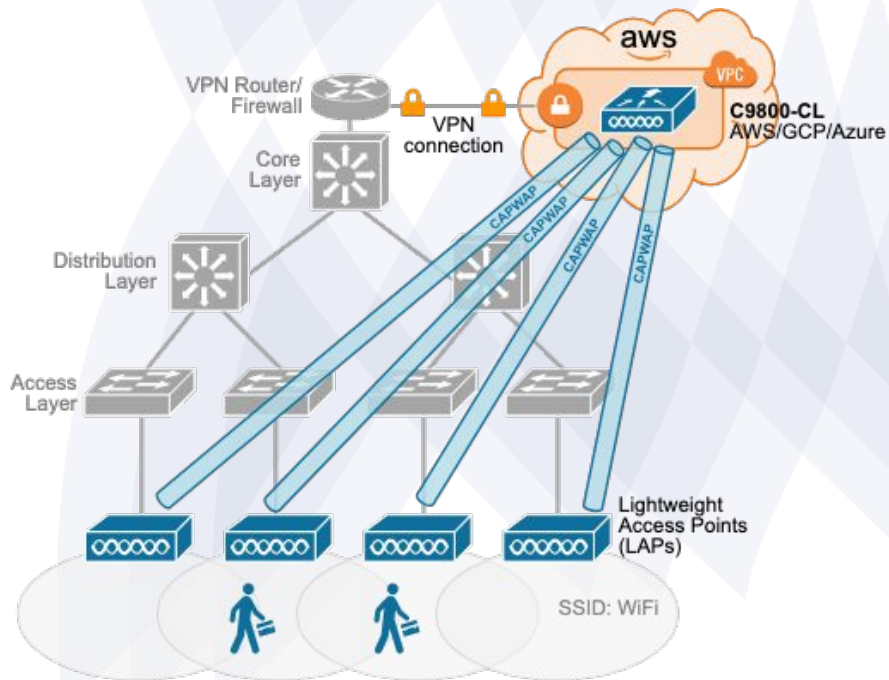


Wireless Controller Placement: WLC in the Cloud



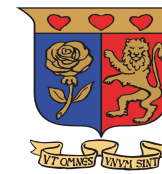
Strathmore
UNIVERSITY

- However, a virtual WLC may introduce latency concerns if deployed in a remote cloud environment far from the access points, potentially affecting roaming and real-time applications. Unlike a hardware-based controller, it relies on virtual infrastructure, which may require more IT expertise to manage, especially in terms of cloud networking and security. Organizations must also consider bandwidth consumption, as all control traffic between APs and the cloud-hosted controller must traverse the network.



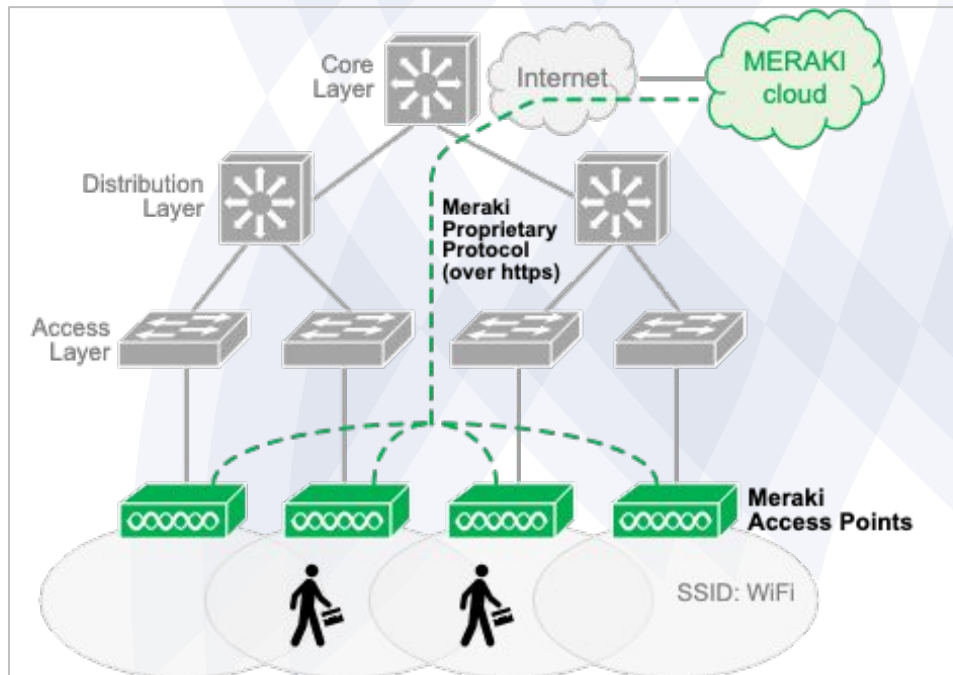
- An on-premises WLC (like the Catalyst 9800 Series) offers lower latency and direct control over the wireless infrastructure, making it more suitable for high-density or mission-critical environments.
- On-prem controllers also provide greater security control, as traffic remains within the organization's network instead of being processed in the cloud. However, they come with higher hardware costs and may require redundant controllers for high availability and more staff to manage the high availability.

Wireless Controller Placement: WLC in the Cloud- Cisco Meraki

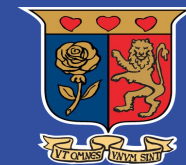


Strathmore
UNIVERSITY

- The fully cloud managed solution a full enterprise WLC that supports advanced RF management, QoS, security, and mobility services, just like a physical WLC.
- It does not require the cloud for management but can be deployed in a cloud environment if desired. However, network administrators still need to configure, update, and manage the controller manually, similar to traditional on-premises WLCs.



- Cisco Meraki Wireless is a 100% cloud-managed solution where all configuration, monitoring, and management are handled through the Meraki Dashboard.
- There is no need for an on-prem or virtual controller, as Meraki APs connect directly to the Meraki cloud for centralized control.
- The trade-off is that Meraki offers less customization and advanced control compared to Catalyst controllers, making it better suited for simpler deployments rather than complex, high-performance enterprise environments.



Strathmore
UNIVERSITY